

# BROADBAND CHAOS GENERATED IN A TIME-DELAYED OPTO-ELECTRONIC DEVICE

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## Abstract

The study of nonlinear dynamics, specifically chaotic dynamics, has grown in popularity since its debut in the 1970s. Today, researchers are interested in practical applications of chaos, such as chaos communications and ranging, which requires simple devices that produce complex and high-speed dynamics. The subject of this thesis is an opto-electronic oscillator comprised of commercially available components which, due to the presence of a feedback loop and nonlinear element, can display a variety of behaviors. By adjusting the gain in the feedback loop and the effective slope of the nonlinearity's transfer function, the dynamics of the system can be tuned to be periodic, quasiperiodic, or chaotic. The chaotic behavior that results at a particular operating point of the nonlinearity is high-speed, with a flat, broad power spectrum extending out to 8 GHz (the cutoff frequency of the oscilloscope used to measure the dynamics). To the best of my knowledge, the "featureless" broadband chaos I observe in this system has not been reported in the literature. In addition, this system's design is relatively simple, making it an attractive device for use in practical applications of chaos.

Furthermore, at the effective slope of the nonlinearity where broadband chaos is observed, a (noise-free) model describing the system shows that the steady-state is linearly stable for all values of the feedback gain. Experimentally, however, I find that the steady-state solution is lost as I increase the gain beyond a certain threshold. As the noise in the system is increased, this threshold lowers further. I hypothesize that the presence of noise in the system pulls the bifurcation threshold at this operating point down from infinity and makes the coexisting chaotic attractor accessible. To test this hypothesis, I investigate the dynamics of a delay differential equation and a one-dimensional map, both written to model the features of the physical system. I find that

a finite perturbation can drive the system towards the chaotic attractor, and provide a criteria for the size of the perturbation necessary to leave the steady-state as a function of the feedback gain.

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plan. Her presence in the department has been greatly missed.

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# Chapter 1

## Introduction

### 1.1 Dynamics of Systems with Nonlinear Time-Delayed Feedback

Ever since the 1970s, when the pioneering work of Edward Lorenz caught the interest of scientists around the world, the study of chaos and complex systems has grown exponentially [1]. Currently, a large number of scientists are devoting their research to developing chaotic devices for use in a variety of applications. The subject of this thesis is one such chaotic device. In order for chaotic dynamics to be possible, however, one needs to have a nonlinear system with at least three dynamical variables (*i.e.*, a three dimensional phase space), which often means the systems themselves must be rather complex [1]. One way around this is to utilize systems with nonlinear time-delayed feedback that obey delay differential equations (DDEs).

A DDE is an equation in which the value of a dynamic variable at a given time depends on the values of other dynamic variables at current and previous times. The phase space corresponding to such an equation of motion is infinite dimensional, allowing for the possibility of chaotic solutions. Therefore, systems that need to be modeled with DDEs can often be comprised of a small number of simple components, and yet can still give rise to complicated dynamics due to their infinite dimensional phase space.

Even though DDEs are used to model the behavior of many types of systems (*i.e.*, physiological diseases, population dynamics, neuronal networks, and nonlinear opti-

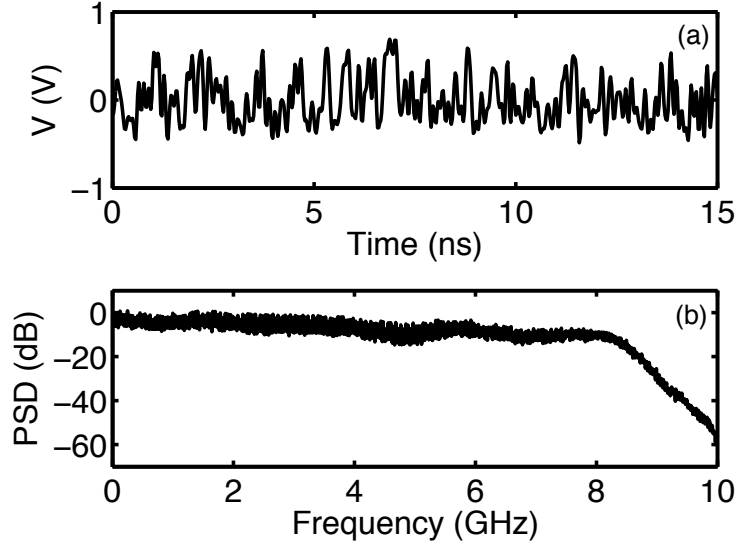
cal devices [2]), the study of these equations is a relatively new and difficult area of mathematical research. One use of my experimental setup, therefore, is to gain insight about the solutions and properties of a particular class of DDEs experimentally. This system's simple design yet complex behavior also make it an attractive device for use in practical applications of chaos, such as chaos communications [3], synchronization [4], and ranging [5]. In addition, there is currently a growing interest among engineers to create ultrastable microwave oscillators [6, 7] and ultrawideband (UWB) devices [8, 9], both of which are possible with my device.

There are also potential applications for chaotic oscillators that have yet to be realized experimentally. For example, several of these devices could be networked in such a way that breaking or altering any of the couplings would lead to a dramatic change in the properties of the network. One potential application of such a network is to create a sensitive, low-profile intrusion detection system [10].

## 1.2 Overview of Thesis

In this thesis, I investigate the dynamics of an opto-electronic oscillator subject to nonlinear time-delayed feedback with four different methods: experimentation with the physical system, linear stability analysis of a noise-free model, numerical simulation of a stochastic model, and analysis of a map derived to approximate certain features of the physical system and full model. I introduce the physical system in Chapter 2, describing the operating principles and characteristics of each component. The chapter culminates with the derivations of both a deterministic and a stochastic model, each including a time-delayed variable, that approximate the system's dynamics. Next, I describe the research others have completed on systems with nonlinear time-delayed feedback in Chapter 3, starting with the pioneering work of Ikeda. This research is

divided into two categories: low-pass feedback and bandpass feedback.



**Figure 1.1:** Time series (a) and power spectrum (b) for the broadband chaos observed in the physical system. The power spectrum is essentially flat up to the cutoff frequency of the measurement device ( $\sim 8$  GHz).

I then present my experimental findings in Chapter 4, the most notable of which is the discovery of high-speed chaotic oscillations with a broad, flat power spectrum, similar to that of white noise, as shown in Fig. 1.1. The transient behavior connecting the steady-state to chaotic dynamics initially takes the form of a sequence of pulses with successively increasing amplitudes and separated in time by the amount of time it takes a signal to transverse the feedback loop. In Chapter 5, I perform a linear stability analysis on the deterministic model and show that the steady-state is linearly-stable for all values of the feedback gain in the regime where I observe the broadband chaos. I then integrate the stochastic model, and find that, when I apply a finite perturbation to the steady-state, I am able to access the coexisting chaotic attractor. This motivates me to model the pulsing behavior seen in both the experiment and numerical simulation with a one-dimensional map. In Chapter 6, I compare the results obtained from each

of these approaches, and find that the one-dimensional map can be used to predict the conditions under which a finite perturbation in either the experiment or simulation will drive the system away from the steady-state. Finally, I conclude with some ideas for possible avenues to pursue in the future.

# Chapter 2

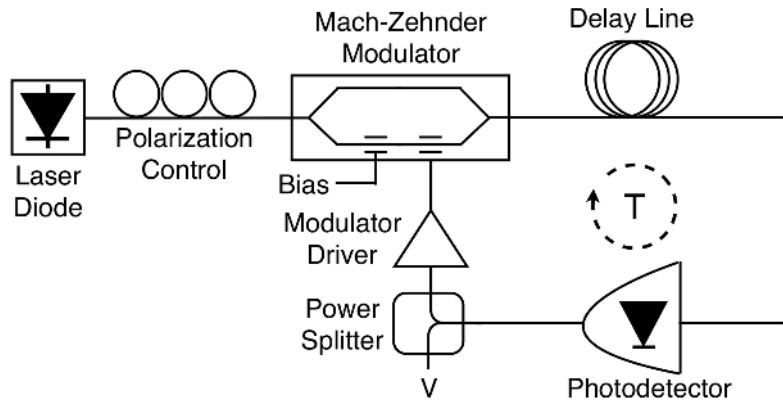
## System Description

The opto-electronic system under investigation is comprised of commercially available components which, due to the presence of a feedback loop and nonlinear elements, displays a variety of behaviors. As shown in Fig. 2.1, the setup is as follows: light with a wavelength of 1550 nm generated in a semiconductor laser propagates through a single mode optical fiber, a polarization controller, and a Mach-Zehnder modulator (MZM). The light exiting the modulator is incident on a photodetector, and the resulting voltage is amplified by a modulator driver and fed back into the MZM via a radio-frequency (RF) input. It is this voltage that exhibits the behavior I am interested in. By adjusting the gain in the feedback loop, the operating point of the nonlinearity, and the length of the time delay, the dynamics of the system can be tuned to be periodic, quasi-periodic, or chaotic. This system is best described by a DDE since the response time is on the order of picoseconds, while the time delay of the feedback loop is on the order of nanoseconds. The following sections describe each of the system components in greater detail.

### 2.1 The Laser Diode

Throughout this section, I am following the development given by Agrawal and Dutta [11] and Petermann [12].

The first semiconductor laser (or laser diode) was created in 1962, just four years after the invention of the laser. Laser diodes have since developed into an essential



**Figure 2.1:** Schematic of the experimental setup showing the key system components and the order in which a signal originating from the laser diode propagates through them. The measured electrical signal  $V$  is taken from one arm of the power splitter.

element of many optoelectronic systems. Lasers of this type can produce radiation with wavelengths anywhere from 0.3 to 100  $\mu\text{m}$ . Due to the low-loss transmission window in optical fibers at 1550 nm, many laser diodes are designed to emit infrared radiation at this wavelength. The Sumitomo InGaAsP/InP distributed-feedback multi-quantum-well laser diode used in my experimental setup (model SEI SLT5411) emits at this wavelength.

### 2.1.1 Operating Principles

All lasers require a gain medium where stimulated emission of photons can occur, a means to supply energy to the gain medium in order to achieve a population inversion, and optical feedback to amplify the signal. In a laser diode, the forward-biased p-n junction supplied with an injection current fulfills all of these criteria. Physically, electrons from the conduction band of the n-type semiconductor material and holes from the valence band of the p-type semiconductor material are forced into the depletion region (due to the applied potential difference) where they can recombine and release

a photon. This can happen via spontaneous emission, where the direction and phase of the photon are random, or stimulated emission, where one photon stimulates the release of an identical photon. In order for lasing to occur, the rate of stimulated emission must be greater than the rate of loss due to absorption, which will happen for high enough current densities in the depletion region. Double heterostructure laser diodes, where light is confined to the innermost p-n junction region by surrounding it with a cladding layer of semiconductor materials that have a higher band gap and lower index of refraction, can therefore have a lower lasing current threshold than laser diodes with a single p-n junction.

As with all lasers, laser diodes will emit radiation with several different frequencies unless certain precautions are taken. The laser used in my experiments is a distributed-feedback (DFB) semiconductor laser and can produce light with nearly a single frequency. One longitudinal mode is selected by varying the thickness of one of the cladding layers periodically along the cavity length. This grating produces backward Bragg scattering, which will result in light with a wavelength that depends on the pitch of the grating. The grating also provides the optical feedback necessary for lasing to occur.

The behavior of a single transverse longitudinal electromagnetic wave (subject to several simplifying assumptions) is governed by the rate equations [11]

$$\dot{P} = (G - \gamma_p)P + R_{sp}, \quad (2.1)$$

$$\dot{N} = \frac{I}{q} - \gamma_e N - GP, \quad (2.2)$$

$$\dot{\phi} = -\frac{\bar{\mu}}{\mu_g}(\omega - \Omega) + \frac{1}{2}\beta(G - \gamma_p). \quad (2.3)$$

Here,  $P$  is the number of photons (proportional to the emitted laser power  $P_{LD}$ ),  $N$  is the number of charge carriers, and  $\phi$  is the phase of the field. One can see that the

rate of photon production depends on the difference between the net rate of stimulated emission (denoted by  $G$ , which is a function of  $P$  and  $N$ ) and the photon decay rate  $\gamma_p$ , as well as the rate of spontaneous emission (denoted by  $R_{sp}$ , which is a function of  $N$ ). The rate of change of the number of charge carriers, on the other hand, increases at the rate at which charges are injected into the active region (given by  $I/q$ , where  $I$  is the current and  $q$  is the charge of one of the carriers), decreases at the carrier-recombination rate (denoted by  $\gamma_e$ , which is a function of  $N$ ), and decreases further when a photon is created due to stimulated emission at a rate of  $GP$ . The rate of change of the phase depends on the mode and group indexes of refraction, denoted by  $\bar{\mu}$  and  $\mu_g$ , respectively, the steady-state frequency  $\omega$ , the cavity resonance frequency  $\Omega$ , and the linewidth enhancement factor  $\beta$ .

### 2.1.2 Steady-State Laser Behavior

The steady-state behavior for constant injection current  $I$  can be determined from Eqs. 2.1-2.3 by setting the derivatives on the left-hand side equal to zero. To do this exactly, the functional dependence of  $R_{sp}$ ,  $\gamma_e$ , and  $G$  on  $N$  and  $P$  needs to be determined. Instead, one can make a few simplifying assumptions and obtain a prediction for the output power as a function of current that agrees reasonably well with experimental data. If one neglects the contribution to the photon number rate due to spontaneous emission (*i.e.*,  $R_{sp} \rightarrow 0$ ), ignores any dependence of  $\gamma_e$  on  $N$ , and assumes that the rate of stimulated emission only depends linearly on  $N$  and is given by  $G = AN - B$ , one obtains the following equations for the steady-state values  $P^*$  and  $N^*$

$$0 = (AN^* - B - \gamma_p)P^*, \quad (2.4)$$

$$0 = \frac{I}{q} - \gamma_e N^* - (AN^* - B)P^*. \quad (2.5)$$

The two fixed-point solutions are then

$$P^* = 0, N^* = \frac{I}{q\gamma_e}, \quad (2.6)$$

and

$$P^* = \frac{I}{q\gamma_p} - \frac{\gamma_e(B + \gamma_p)}{A\gamma_p}, N^* = \frac{B + \gamma_p}{A}. \quad (2.7)$$

Linear stability analysis shows that, for currents below

$$I_{th} = q\gamma_e \frac{B + \gamma_p}{A}, \quad (2.8)$$

the first fixed point (Eq. 2.6) is stable, which corresponds to a steady-state power of zero. Above  $I_{th}$ , one finds that the second fixed point (Eq. 2.7) is stable, which corresponds to a steady-state power that increases linearly with increasing current

$$P^* = \frac{1}{q\gamma_e}(I - I_{th}). \quad (2.9)$$

In Chapter 4, I verify the above model by experimentally determining the dependence of the laser's output power on the injection current.

### 2.1.3 Noise Characteristics

The previous subsection ignored the possibility of intensity and phase noise due to the quantum nature of the photons and carriers involved in the lasing process. One can treat the fluctuations caused by spontaneous emission and carrier-generation-recombination processes by adding corresponding Langevin noise sources to the rate equations. Following the development given in [11], this gives the following stochastic

rate equations

$$\dot{P} = (G - \gamma_p)P + R_{sp} + F_p(t), \quad (2.10)$$

$$\dot{N} = \frac{I}{q} - \gamma_e N - GP + F_N(t), \quad (2.11)$$

$$\dot{\phi} = -\frac{\bar{\mu}}{\mu_g}(\omega - \Omega) + \frac{1}{2}\beta(G - \gamma_p) + F_\phi(t), \quad (2.12)$$

where  $F_p$  and  $F_\phi$  are due to spontaneous emission and  $F_N$  is due to the carrier-generation-recombination process. Note that Eqs. 2.10–2.12 are identical to Eqs. 2.1–2.3, with the exception of the noise terms. If one further assumes that the noise processes are Markovian (*i.e.*, that the process is “memoryless” because the correlation time of the noise sources is much less than the relaxation times  $\gamma_p^{-1}$  and  $\gamma_e^{-1}$ ), then the Langevin noise sources have the following two properties

$$\langle F_i(t) \rangle = 0, \quad (2.13)$$

$$\langle F_i(t)F_j(t') \rangle = 2D_{ij}\delta(t - t'), \quad (2.14)$$

where the  $D_{ij}$  terms are diffusion coefficients and the angled brackets denote ensemble averages. These diffusion coefficients can be obtained by calculating the second moments of  $P$ ,  $N$ , and  $\phi$  from the stochastic rate equations or by physical arguments.

To analyze the noise characteristics, one can perturb  $P^*$ ,  $N^*$ , and  $\phi^*$  by small amounts  $\delta P$ ,  $\delta N$ , and  $\delta \phi$ , and linearize the resulting stochastic rate equations to

obtain

$$\delta \dot{P} = -\Gamma_p \delta P + \left( \frac{\partial G}{\partial N} P + \frac{\partial R_{sp}}{\partial N} \right) \delta N + F_p(t), \quad (2.15)$$

$$\delta \dot{N} = -\Gamma_N \delta N - \left( G + \frac{\partial G}{\partial P} \right) \delta P + F_N(t), \quad (2.16)$$

$$\delta \dot{\phi} = -\frac{1}{2} \beta \frac{\partial G}{\partial N} \delta N + F_\phi(t). \quad (2.17)$$

Here, it should be understood that all of the dynamic variables and their derivatives are evaluated at the second fixed point (Eq. 2.7). The small-signal decay rates of the photon and carrier populations are given by

$$\Gamma_p = \frac{R_{sp}}{P} - P \frac{\partial G}{\partial P}, \quad (2.18)$$

$$\Gamma_N = \gamma_e + N \frac{\partial \gamma_e}{\partial N} + P \frac{\partial G}{\partial N}, \quad (2.19)$$

respectively. By taking the Fourier transform of Eqs. 2.15-2.16, one can solve for the Fourier components of the fluctuations. For example, the fluctuation in photon number (proportional to the intensity noise) at a frequency  $\omega$  is given by

$$\delta \tilde{P}(\omega) = \frac{(\Gamma_N + i\omega) \tilde{F}_p + \left( \frac{\partial G}{\partial N} P + \frac{\partial R_{sp}}{\partial N} \right) \tilde{F}_N}{(\Omega_R + \omega - i\Gamma_R)(\Omega_R - \omega + i\Gamma_R)}, \quad (2.20)$$

where  $\Omega_R$  is called the relaxation oscillation frequency and will be discussed later.

Ultimately, I am interested in the intensity noise, which is typically quantified in terms of the relative intensity noise (RIN). The output of the laser can be described by

$$P_{LD}(t) = \overline{P_{LD}} + \delta P_{LD}(t), \quad (2.21)$$

where  $\overline{P_{LD}}$  is the mean steady-state power and  $\delta P_{LD}(t)$  is a noise term with a time

average of zero, then in terms of the spectral density  $S(\omega)$  of the noise term (as defined in Eq. A.3) the RIN is given by

$$\text{RIN} = \frac{S(\omega)}{\overline{P_{LD}}^2}, \quad (2.22)$$

$$= \lim_{t \rightarrow \infty} \frac{1}{T} \frac{\langle |\delta \tilde{P}_{LD}(\omega)|^2 \rangle}{\overline{P_{LD}}^2}. \quad (2.23)$$

Using Eq. 2.20 and the values for  $D_{PP}$  and  $D_{NN}$ , the RIN becomes

$$\text{RIN} = \frac{2R_{\text{sp}} \left[ \left( \Gamma_N^2 + \omega^2 \right) + \frac{\partial G^2}{\partial N} P^2 \left( 1 + \frac{\gamma_e N}{R_{\text{sp}} P} \right) \right]}{P \left[ \left( \Omega_R^2 - \omega^2 \right)^2 + (2\omega \Gamma_R)^2 \right]}. \quad (2.24)$$

One can immediately see that a resonance occurs at the relaxation oscillation frequency, which will correspond to a peak in the power spectrum near  $\Omega_R$  for a constant laser power. Relaxation oscillations can be explained physically by considering how the photon number and population inversion are related to one another: as the photon number increases, the population inversion decreases due to the increased rate of stimulated emission transitions, leading to a decrease in the gain, which results in an increase in intensity [13]. The steady-state behavior is approached via damped oscillations at  $\Omega_R$ , which is typically between 1 and 10 GHz for most semiconductor lasers [14].

Additionally, for  $\omega \ll \Omega_R$ , the RIN decreases with laser power as  $P^{-3}$ , and for large  $\omega$  it goes as  $P^{-1}$ . (To show this, one has to assume that  $\Omega_R$  is proportional to  $P^{1/2}$ .) Thus, only for frequencies well below  $\Omega_R$  and for constant laser power, can I assume an additive noise term  $\delta P_{LD}(t)$  with a white frequency spectrum in Eq. 2.21.

The RIN for the laser in my experiment in this regime is quoted as -140 dB/Hz by the manufacturer (where the units dB/Hz arise because they have taken the logarithm

of a number with units of 1/Hz). The spectral density can be computed from

$$S(\omega) = 10^{\text{RIN}/10} \overline{P_{LD}^2}, \quad (2.25)$$

$$= 10^{-14} [\text{s}] \overline{P_{LD}^2}. \quad (2.26)$$

I can then approximate the mean-square power fluctuations by

$$\overline{\delta P_{LD}^2} = 2 \int_{\omega_{hp}}^{\omega_{lp}} S(\omega) d\omega, \quad (2.27)$$

$$= 2 \times 10^{-14} [\text{s}] (\omega_{lp} - \omega_{hp}) \overline{P_{LD}^2}, \quad (2.28)$$

where  $\omega_{hp}$  and  $\omega_{lp}$  are the high- and low-pass angular frequencies of the photodetector, to be discussed in Sec. 2.4. The manufacturer claims  $\omega_{hp} = 1.88 \times 10^5$  Hz and  $\omega_{lp} = 8.17 \times 10^{10}$  Hz. Thus, for my system,

$$\frac{\overline{\delta P_{LD,rms}}}{\overline{P_{LD}}} \approx 4.0\%. \quad (2.29)$$

## 2.2 The Polarization Controller

The MZM (described in the next section) is a polarization sensitive device and will only function properly for light linearly polarized along a particular direction. The light exiting the laser diode is linearly polarized, but the polarization can change along the optical fiber due to birefringence. This can be understood by noting that the index of refraction of glass decreases when it is compressed and increases when it is expanded [15]. Bending the fiber compresses the glass one direction and expands it in another, which induces birefringence and leads to a change in the polarization state.

The polarization controller I use consists of a quarter-waveplate, half-waveplate,

and another quarter-waveplate connected in series. The waveplates are constructed by simply looping fiber around a spool (with the half-waveplate having twice as many loops as the quarter-waveplates) to compress the glass in the direction parallel to the plane of the spool and expanding it in the plane perpendicular to the spool, thus inducing birefringence. The quarter-waveplates introduce a  $\pi/4$  phase shift between the two polarization axes and convert linearly polarized light into elliptically polarized light or vice versa. The half-waveplate introduces a  $\pi/2$  phase shift, which causes the direction of polarization to flip about the fast axis. By manually rotating the fast axis of each of the waveplates, I can adjust the polarization state of the light exiting the polarization controller and incident on the MZM.

## 2.3 The Mach-Zehnder Modulator (MZM)

The MZM is a 10 Gb/s Integrated Optic Intensity Modulator. It modulates the intensity of an incident optical signal by exploiting Pockels electrooptic effect in a Lithium Niobate crystal situated in one arm of a Mach-Zehnder interferometer. The Pockels effect causes the index of refraction for a particular polarization state to depend linearly on the applied electric field [15]. By splitting an optical signal with a 50/50 splitter and passing one beam through the crystal, which is sandwiched by a parallel plate capacitor, one can control the phase difference between the two beams. To control the phase difference, one simply varies the voltage across the capacitor, which alters the optical path length for one of the beams. Upon recombination, the resulting optical signal can have an intensity anywhere from zero up to the intensity of the incoming signal (assuming there is no insertion loss and that the incoming light has the correct polarization).

To find the transmission function for the MZM, I write the complex electric field at

the output of the modulator as a combination of two fields with equal amplitudes and different wave numbers ( $k_1$  and  $k_2$ )

$$E = \frac{E_0}{2}e^{ik_1L} + \frac{E_0}{2}e^{ik_2L}, \quad (2.30)$$

where  $L$  is the length of each arm of the interferometer. If I then factor out  $e^{ik_1L/2}e^{ik_2L/2}$  from both terms to obtain

$$E = \left( \frac{E_0}{2}e^{ik_1L/2}e^{-ik_2L/2} + \frac{E_0}{2}e^{-ik_1L/2}e^{ik_2L/2} \right) e^{ik_1L/2}e^{ik_2L/2}, \quad (2.31)$$

$$= E_0 \cos \Delta k L e^{i\bar{k}L}, \quad (2.32)$$

where  $\Delta k = (k_1 - k_2)/2$  and  $\bar{k} = (k_1 + k_2)/2$ . Because the intensity is proportional to the square of the electric field, I find that

$$I = I_0 \cos^2 \left( \frac{\pi V_{mod}}{2V_\pi} \right), \quad (2.33)$$

where  $V_{mod}$  is the total voltage across the capacitor (minus the voltage  $V_{max}$  that yields maximum transmission) and  $V_\pi$  is the value of the voltage across the capacitor when the intensity is zero.

There are two methods of adjusting  $V_{mod}$  and, hence, the transmission function of the MZM: a constant bias voltage  $V_B + V_{max}$  provided by an external power supply and a time-varying voltage  $V_{MD}$  provided by modulator driver in the feedback loop. Each port has a characteristic  $V_\pi$ , denoted  $V_{\pi,DC}$  and  $V_{\pi,RF}$ . The output power  $P_{MZ}$  of the MZM displays a typical interference pattern as a function of the total applied voltage

given by

$$P_{MZ} = g_{MZ} P_{LD} \cos^2 \left[ \frac{\pi}{2} \left( \frac{V_B}{V_{\pi,DC}} + \frac{V_{MD}}{V_{\pi,RF}} \right) \right], \quad (2.34)$$

where  $g_{MZ}$  is the insertion loss and  $P_{LD}$  is the power incident on the MZM from the laser diode. In Chapter 4, I will explain how  $V_{max}$ ,  $V_{\pi,DC}$ ,  $V_{\pi,RF}$ , and  $g_{MZ}$  are determined.

## 2.4 The Photodetector

To convert the optical signal into an electrical signal, I use an optical receiver manufactured by Miteq (model DR-125G). The optical signal is coupled to the photodiode via a single-mode optical fiber, making it ideal for my purposes. In the following subsections, I follow the development given by Boyd [16].

### 2.4.1 Operating Principles

Like laser diodes, photodiodes are essentially p-n junctions with an applied potential difference. Unlike laser diodes, however, the junction is typically reversed biased so that the width of the depletion region is extended. When radiation within a particular frequency range is incident on this region, electron-hole pairs are created and swept out of the region in opposite directions due to the external bias. This photocurrent is proportional to the intensity of the light. Additional contributions to the current are due to diffusion of both positive and negative charge carriers into the depletion region ( $i_{pd}$  and  $i_{nd}$ ) and thermal excitation that mobilizes both positive and negative charge carriers that are mobilized due to thermal excitations ( $i_{pg}$  and  $i_{ng}$ ). The diffusion current depends on the magnitude of the bias, whereas the thermally generated current is independent of it. Accounting for the directions each type of charge will move under

the influence of the reverse bias, the total current is given by

$$i = i_{pd} + i_{nd} - i_{pg} - i_{ng} - \frac{\eta e P}{h\nu}, \quad (2.35)$$

where  $\eta$  is the quantum efficiency of the detector,  $e$  is the charge of an electron,  $P$  is the optical power incident on the detector,  $h$  is Planck's constant, and  $\nu$  is the frequency of the incident radiation. As one can see, if the current  $i$  is passed through a resistor, the voltage across the resistor will depend linearly on the incident optical power.

### 2.4.2 Response and Noise Characteristics

Photodiodes are characterized by four main quantities: responsivity, bandwidth, gain, and noise. The responsivity of the detector is defined as the ratio of the output signal to the input power and is wavelength dependent. For the receiver used in this experiment, the responsivity  $\mathcal{R}$  is quoted as 0.9 A/W at 1550 nm by the manufacturer. The responsivity also depends on the modulation frequency. The range of frequencies for which the responsivity is within 3 dB of its maximum value is defined as the bandwidth of the device, and, for my detector, this frequency range is said to be 30 kHz to 13 GHz by the manufacturer. Because, in the end, I am interested in the voltage produced by this device, I need to know its optical-to-electrical transfer characteristics. I call this quantity the detector gain  $g_{PD}$  and will show later that it is approximately -3.2 V/mW, where the negative sign indicates that the signal is inverted. Ignoring, for the moment, the relatively small contribution to the voltage due to diffusion and thermal excitation, an optical signal exiting the MZM with a power  $P_{MZ}$  will produce a photodiode voltage of

$$V_{PD} = g_{PD} P_{MZ}. \quad (2.36)$$

Equation 2.36 ignores the noise sources present in the system. The two main contributions to the noise are due to the discrete nature of the charge carriers in the current (shot noise) and thermal fluctuations in the resistor (Johnson Noise).

The fact that the (discrete) charge carriers pass through the resistor with an average rate  $r$ , but at random times, gives rise to statistical fluctuations in the output voltage. This shot noise can be analyzed with the Poisson distribution. Thus, the probability  $p(N)$  that  $N$  charge carriers will pass through in a time interval  $T$  is given by

$$p(N) = \frac{\bar{N}^N}{N!} e^{-\bar{N}}, \quad (2.37)$$

where  $\bar{N} = rT$  is the average number of “events” occurring in  $T$ . The average current is then

$$\bar{i} = \frac{e\bar{N}}{T}. \quad (2.38)$$

One way to characterize the noise in the current is to calculate its variance

$$\overline{i^2} - \bar{i}^2 = \frac{e\bar{i}}{T}, \quad (2.39)$$

$$= 2e\bar{i}\Delta f, \quad (2.40)$$

where I have used the fact that the variance of a Poisson random variable is equal to its mean and that the bandwidth of the detection system is related to the averaging time by  $\Delta f = 1/(2T)$ . Furthermore, if one denotes the shot noise in the voltage due to the current as  $\delta V_s$ , the variance of the current is proportional to its mean-square

amplitude

$$\overline{\delta V_S} \propto 2e\bar{i}\Delta f. \quad (2.41)$$

Each of the contributions to the current given in Eq. 2.35 are statistically independent sources of shot noise, which upon addition in quadrature gives

$$\overline{\delta V_S^2} \propto e \left( \overline{i_{pg}} + \overline{i_{pd}} + \overline{i_{ng}} + \overline{i_{nd}} + \frac{\eta \bar{P}}{h\nu} \right) \Delta f. \quad (2.42)$$

Equation 2.40 is known as Schottky's formula and noise processes that take this form are referred to as shot noise. Notice that if one divides through both sides of Eq. 2.40 by the bandwidth, then this shows that the average noise power per unit frequency (*i.e.*, the spectral density) is independent of frequency. Thus, the discreteness of the charge carriers gives rise to a source of white noise in the current (and hence the output voltage) of the photodetector.

An additional voltage fluctuation (called Johnson noise) arises when electrons experience thermal fluctuations in a resistor. Since our detector can be thought of as a current source in series with a resistor at a finite temperature, it is important to understand the characteristics of the noise in the voltage across this resistor. If one considers a parallel combination of a resistor  $R$  and capacitor  $C$  in equilibrium at temperature  $T$ , then the thermal agitation of electrons in this system will produce a fluctuating voltage  $\delta V_J$ . Since the only degree of freedom in this system is the voltage across one of the elements, the equipartition theorem requires that

$$\frac{1}{2} C \overline{\delta V_J^2} = \frac{1}{2} k_B T, \quad (2.43)$$

where  $k_B$  is Boltzman's constant. The one-sided spectral density for Johnson noise

can be computed if the autocorrelation function  $C(\tau)$  is known. Assuming a transient decay time constant of  $RC$ ,

$$C(\tau) = \overline{\delta V_J(t) \delta V_J(t + \tau)}, \quad (2.44)$$

$$= \overline{\delta V_J(t) \delta V_J(t)} e^{-\tau/RC}, \quad (2.45)$$

$$= \overline{\delta V_J^2} e^{-\tau/RC}. \quad (2.46)$$

Using Eq. A.10, the one-sided spectral density becomes

$$W(\omega) = \frac{4\overline{\delta V_J^2} RC}{1 + (\omega RC)^2}, \quad (2.47)$$

$$\approx 4\overline{\delta V_J^2} RC, \quad (2.48)$$

where the last step is true if  $\omega \ll RC$ . Finally, from Eq. 2.43, it is clear that  $\overline{\delta V_J^2} = k_B T/C$ , which gives

$$W(\omega) \approx 4k_B TR. \quad (2.49)$$

Notice that the one-sided spectral density derived here is independent of  $C$ . Thus, if one takes the limit that  $C \rightarrow 0$ , then  $\omega \ll RC$  will be valid for all values of  $\omega$  and  $R$  and the approximation holds. The spectral density is also independent of frequency, making this another source of white noise.

I have shown that both dominant noise sources can be treated as frequency independent, but only the Johnson noise is independent of the incident radiation power. Thus, for high enough powers, the shot noise will dominate and we expect the root-mean-square amplitude of the noise to scale as  $P^{1/2}$ . For steady-state operation at

constant  $P_{LD}$ , I can write the output voltage of the photodetector as

$$V_{PD} = g_{PD}P_{MZ} + \delta V_{PD}, \quad (2.50)$$

where the noise term  $\delta V_{PD}$  is characterized by a white frequency spectrum and a constant mean-square amplitude  $\overline{\delta V_{PD}^2(t)}$ . With no incident power from the laser, I record a long time series at a sampling rate of 40 GHz with a digital oscilloscope (Agilent Infinium DSO80804B) having a bandwidth of 8 GHz. With this data, I measure  $\delta V_{PD,rms}$  to be approximately 2.16 mV<sub>rms</sub>

## 2.5 The Bandpass Filter

The electronics that comprise the system are bandpass filtered by the inherent bandwidth limits of each device. For simplicity, I model the entire electronic portion as if there is one high-pass and one low-pass corner frequency (*i.e.*, a two-pole bandpass filter placed at the output of the photodetector). The transfer function for such a filter with angular bandwidth  $\Delta$  and angular frequency of maximum transmission  $\omega_0$  can be expressed in the frequency domain as

$$H(s) = \frac{\Delta s}{s^2 + \Delta s + \omega_0^2}, \quad (2.51)$$

where  $s = i\omega$ . In terms of the high-pass and low-pass corner frequencies,  $\Delta = 2\pi(f_+ - f_-)$  and  $\omega_0^2 = (2\pi)^2 f_+ f_-$ .

By definition, the transfer function is the ratio of the output signal to the input signal in the frequency domain. I am interested, however, in how the input and output to the filter relate in the time domain. It turns out that, in the time domain, the

bandpass-filtered signal  $V_{BP}$  is given by

$$V_{BP} + \frac{1}{\Delta} \frac{dV_{BP}}{dt} + \frac{\omega_0^2}{\Delta} \int_0^t V_{BP}(l) dl = V_{PD}, \quad (2.52)$$

where  $V_{PD}$  is the input voltage to the bandpass filter. To verify that this is equivalent to Eq. 2.51, one can simply take the Laplace transform of Eq. 2.52 and recover  $H(s)$ .

## 2.6 The Modulator Driver

I use a 10 Gb/s JDSU optical modulator driver (model H301) to amplify the RF signal used to drive the MZM. The driver has a bandwidth ranging from 75 kHz to 10 GHz and a nonlinear response - the response saturates at high drive voltage. In greater detail, for sinusoidal inputs with low amplitude, the output of the driver will also be sinusoidal. As the amplitude of the input is increased, the driver saturates and the output begins to square off. I will show in Chapter 4 that, in terms of the voltage at the output of the filter  $V_{BP}$ , the signal amplified by the modulator driver  $V_{MD}$  is approximately given by

$$V_{MD} = V_{sat} \tanh \left[ \frac{g_{S1} g_{MD} V_{BP}}{V_{sat}} \right], \quad (2.53)$$

where  $g_{S1}$  is the insertion loss in the arm of the power splitter (Picosecond Pulse Labs 6 dB Power Divider 5331) that feeds into the driver,  $g_{MD}$  is the linear gain of the driver (which is a negative quantity since the amplifier is inverting), and  $V_{sat}$  is the saturation voltage of the driver. I will report how I determine these quantities in Chapter 4.

The voltage produced by the driver will fluctuate due to the noise contained in the input signal and the noise added by the amplification process [17]. Similar to the photodetector, the two main amplifier noise sources will be shot noise, due to the

discrete nature of the electrons, and Johnson noise, due to thermal fluctuations. To get an idea of the magnitude of this noise, I input the noise generated by the Miteq photodetector with no incident optical power to the power splitter and attach one of its arms to the driver. I then measure the resulting noise from the driver  $\delta V_{MD}$  with the 8 GHz digital oscilloscope and find it to be  $43 \text{ mV}_{\text{rms}}$ , which is a factor of 22 greater than the noise measured from the other arm of the power splitter. Since the power splitter is highly symmetric, the noise I measure from one of its arms should be nearly equal to the noise entering the driver. As I will show in Chapter 4,  $g_{MD} = 22.6$ , which indicates that a negligible amount of noise is added to the signal by the driver, rather the input noise is just amplified by approximately  $g_{MD}$ .

## 2.7 Deterministic Model

To model the opto-electronic system one needs to consider: the nonlinear transmission functions of the MZM and modulator driver, the finite bandwidth of the system components, and the amount of time it takes the signal to propagate from the output of the MZM back to the RF input of the MZM. Since the time it takes a signal to transverse the feedback loop ( $T \sim 24 \text{ ns}$ , as I will show in Chapter 4) is on the same order as the timescales of its fluctuations ( $f \sim \text{GHz}$ ), I need to account for this delay by modeling my system with a delay-differential equation (DDE). In contrast to systems without delay, the phase space for time-delayed systems can be infinite dimensional. One can understand this by noting that the dimension of phase space is equal to the number of initial conditions you must specify to determine a solution. For time-delayed systems, one needs to specify an infinite number of initial conditions to uniquely determine a solution (*i.e.*, one must specify the dependent variable over a range of times equal to the length of the delay). In regards to my system, this means that the voltage modulat-

ing the optical signal in one arm of the MZM originated from the optical signal present a time  $T$  earlier (where  $T$  is the length of the time delay).

By combining the effects of the transmission functions, bandpass characteristics, and time-delay, I arrived at the following integro-differential delay equation for the measured voltage  $V$

$$V(t) + \frac{1}{\Delta} \frac{dV(t)}{dt} + \frac{\omega_0^2}{\Delta} \int_0^t V(l) dl = P_{LD} g_{MZ} g_{S0} g_{PD} g_{S2} \cos^2 \left\{ \frac{\pi V_B}{2V_{\pi,DC}} + \frac{\pi V_{sat}}{2V_{\pi,RF}} \tanh \left[ \frac{g_{S1} g_{MD} V(t-T)}{g_{S2} V_{sat}} \right] \right\}. \quad (2.54)$$

Here,  $g_{S0}$  is the insertion loss of an optical splitter placed after the MZM,  $g_{S2}$  is the insertion loss in the arm of the power splitter that connects to the measuring device, and all other variables have been previously defined, with values given in Table 2.1. I will explain how I determine each of the quantities that enter Eq. 2.54 in Chapter 4.

To clean up the notation, I define  $G = P_{LD} g_{MZ} g_{S0} g_{PD} g_{S2}$  and  $g = g_{S1} g_{MD} / g_{S2} V_{sat}$  so that the model can be written more compactly as

$$V(t) + \frac{1}{\Delta} V'(t) + \frac{\omega_0^2}{\Delta} \int_0^t V(l) dl = G \cos^2 \left\{ \frac{\pi V_B}{2V_{\pi,DC}} + \frac{\pi V_{sat}}{2V_{\pi,RF}} \tanh [gV(t-T)] \right\}, \quad (2.55)$$

where the prime indicates a derivative with respect to physical time  $t$ .

To render the equation dimensionless, I rescale time with the bandwidth  $\Delta$

$$s = \Delta t, \quad (2.56)$$

**Table 2.1:** The values of the experimental parameters.

| Description                               | Symbol       | Value                 | Units |
|-------------------------------------------|--------------|-----------------------|-------|
| Filter high-pass frequency                | $f_-$        | 23.4                  | kHz   |
| Filter low-pass angular frequency         | $f_+$        | 12                    | GHz   |
| Filter bandwidth                          | $\Delta$     | $7.54 \times 10^{10}$ | rad/s |
| Angular frequency of maximum transmission | $\omega_0$   | $1.05 \times 10^8$    | rad/s |
| Laser power                               | $P_{LD}$     | 0 – 20                | mW    |
| Maximum transmission of MZM               | $g_{MZ}$     | 0.23                  | -     |
| Optical splitter insertion loss           | $g_{S0}$     | 0.67                  | -     |
| Photodetector gain                        | $g_{PD}$     | -3.20                 | V/mW  |
| Electronic splitter insertion loss        | $g_{S1}$     | 0.50                  | -     |
| MD gain                                   | $g_{MD}$     | -22.6                 | -     |
| Electronic splitter insertion loss        | $g_{S2}$     | 0.50                  | -     |
| MZ bias voltage                           | $V_B$        | -10 – 10              | V     |
| MD saturation voltage                     | $V_{sat}$    | 9.65                  | V     |
| MZM DC port $\pi$ voltage                 | $V_{\pi,DC}$ | 7.68                  | V     |
| MZM RF port $\pi$ voltage                 | $V_{\pi,RF}$ | 7.40                  | V     |
| Time delay                                | $T$          | 24.1                  | ns    |

**Table 2.2:** The values of the dimensionless parameters.

| Description             | Symbol     | Value                 |
|-------------------------|------------|-----------------------|
| Filter parameter        | $\epsilon$ | $1.95 \times 10^{-6}$ |
| Feedback gain           | $\gamma$   | 0 – 5                 |
| Time delay              | $\tau$     | 1820                  |
| MZ operating point      | $m$        | $0.205 \times V_B$    |
| MD saturation parameter | $d$        | 2.05                  |

define five additional dimensionless parameters (also given in Table 2.2)

$$\epsilon = \omega_0^2 / \Delta^2, \quad (2.57)$$

$$\gamma = P_{LD} g_{MZ} g_{S0} g_{PD} g_{S1} g_{MD} / V_{sat}, \quad (2.58)$$

$$m = \pi V_B / 2V_{\pi,DC}, \quad (2.59)$$

$$d = \pi V_{sat} / 2V_{\pi,RF}, \quad (2.60)$$

$$\tau = \Delta T, \quad (2.61)$$

and make the following substitutions

$$x = \frac{g_{S1} g_{MD}}{g_{S2} V_{sat}} V, \quad (2.62)$$

$$y = \frac{\omega_0^2 g_{S1} g_{MD}}{\Delta g_{S2} V_{sat}} \int V dt - \gamma \cos^2 m. \quad (2.63)$$

After some algebraic manipulation, I arrive at the following dimensionless system of equations

$$\dot{x}(s) = -x(s) - y(s) + \gamma \cos^2 \{m + d \tanh[x(s - \tau)]\} - \gamma \cos^2 m, \quad (2.64)$$

$$\dot{y}(s) = \epsilon x(s), \quad (2.65)$$

where the dots represent derivatives with respect to dimensionless time  $s$ . The definitions and numerical values of the dimensionless variables corresponding to the system under investigation are given in Table 2.2.

## 2.8 Stochastic Model

As I have shown throughout this chapter, noise is added to the system in three of its components: the laser diode, the photodetector, and the modulator driver. If I were to fully incorporate all of these noise sources, the model would be extremely complex and difficult to analyze. Instead, I will make some approximations to the voltage at the input of the bandpass filter in order to derive a simplified stochastic model valid for small  $V$ . I chose this regime because I am intersted in how the steady-state ( $V = 0$ ) loses stability.

Starting with the physical model given by Eq. 2.55, I rewrite the right-hand side in terms of the dimensionless variables  $m$  and  $d$  and Taylor-series-expand the nonlinearity about the  $V = 0$  steady-state solution to give

$$V_{in}(t) \approx G \left[ \cos^2 m - dg \sin(2m)V(t - T) \right]. \quad (2.66)$$

Next, I add the three small white-noise terms discussed previously to yield

$$V_{in}(t) \approx [G + \delta G(t)] \left\{ \cos^2 m - dg \sin(2m) [V(t - T) + \delta V_{MD}(t)] \right\} + \delta V_{PD}(t), \quad (2.67)$$

which can be expanded as

$$\begin{aligned} V_{in}(t) \approx & G \left[ \cos^2 m - dg \sin(2m)V(t - T) \right] - dgG \sin(2m)\delta V_{MD}(t) + \cos^2 m\delta G(t) \\ & - dg \sin(2m)\delta G(t)V(t - T) - dg \sin(2m)\delta G(t)\delta V_{MD}(t) + \delta V_{PD}(t). \end{aligned} \quad (2.68)$$

For sufficiently small  $V$ , the multiplicative noise term can be neglected. By combining

all other noise terms into one variable  $N(t)$ , I can rewrite this as

$$V_{in}(t) \approx G [\cos^2 m - dg \sin(2m)V(t - T)] + N(t). \quad (2.69)$$

But this is just the Taylor series expansion of the nonlinearity about  $V = 0$  with an additive noise term. So, for sufficiently small  $V$ , a first approximation of a stochastic model for the physical system is

$$V(t) + \frac{1}{\Delta} V'(t) + \frac{\omega_0^2}{\Delta} \int_0^t V(l) dl \approx G \cos^2 \{m + d \tanh [gV(t - T)]\} + N(t), \quad (2.70)$$

where  $N(t)$  is a white-noise source whose variance depends  $d$ ,  $g$ ,  $G$ ,  $m$ , and the characteristics of the three previously discussed noise sources. This is the model I will use to simulate the system's behavior.

## 2.9 Summary

In this chapter, I developed a physical model for the system under investigation, using the operating principles of each element as a starting point. Initially, I ignore the effects of noise and develop a model (Eq. 2.54) that takes into account the transmission characteristics of each component and the delay time. I then rewrite this model as a system of two dimensionless equations (Eqs. 2.64 and 2.65) characterized by five dimensionless parameters:  $\epsilon$ , which depends on the properties of the bandpass filter;  $m$ , which is proportional to the operating point set by  $V_B$ ;  $d$ , which characterizes the saturation of the modulator driver;  $\gamma$ , which is the normalized gain of a signal in the feedback loop; and  $\tau$ , which is the dimensionless time-delay.

Later I refined this model (Eq. 2.70) to incorporate the noise inherent in three of the system's components (the laser diode, the photodetector, and the modulator driver).

I give approximate values for each source, argue that all three sources of noise have an approximately white frequency spectrum, and, in the simplified the model (valid for small  $V$ ), I combine the effects of these three sources into one additive white-noise term  $N(t)$ .

For my work, I am interested in the dynamics corresponding to various values of  $m$  and  $\gamma$ , with the values of  $\epsilon$ ,  $d$ , and  $\tau$  held fixed. Of particular interest is the behavior that results when  $m = 0$ .

# Chapter 3

## Previous Work

Time-delayed systems with nonlinear feedback, such as the one described in the previous chapter, have been studied in two different capacities. In the first investigations of such devices, the feedback was low-pass filtered (or at least approximated as such), meaning that frequency components from DC up to some cutoff frequency were allowed to propagate through the feedback loop. More recently, researchers have been interested in bandpass filtered feedback, where the DC component of the signal in the feedback loop is blocked. The inclusion of a high-pass filter has led to some interesting features that are not present in purely low-pass feedback. The results from both types of research are discussed in the following sections.

### 3.1 Low-pass Feedback

Ikeda was the first scientist to study a nonlinear system subject to time-delayed feedback [18]. As previously discussed, these types of systems are interesting because they can be extremely simple and still exhibit rich behavior due to the nonlinearity and time delay. The system Ikeda proposed consisted of a ring cavity with a dielectric material in one arm of the ring and was illuminated continuously by a light source with constant amplitude. The finite propagation time necessary for light to transverse the loop as well as its nonlinear interaction with the dielectric material allowed for the discovery of new types of instabilities. This system motivated Ikeda and his collaborators to

broaden their study to include all DDEs of the form [19, 20]

$$\dot{x}(s) = -x(s) + F[\gamma; x(s - \tau)]. \quad (3.1)$$

They rewrote this DDE as a difference equation and showed that, as you increase the bifurcation parameter  $\gamma$ , the system will transition from steady-state behavior, to periodic behavior, to non-periodic (chaotic) behavior. Additionally, they found that several stable periodic states could coexist for the same parameter values and suggested that these multistable modes could be used to make a dynamic memory. It is important to note that this behavior is not present in a differential equation that does not account for the time-delay.

Shortly after Ikeda's proposal, Gibbs *et al.* [21] reported the first experimental observation of the Ikeda instability in a modified hybrid optically bistable device with time-delayed feedback. They found regimes of periodic behavior with periods of two and four times the length of the time delay (on the order of milliseconds), in addition to chaotic regimes. These results agree well with Ikeda's computations.

The first report of a device using components similar to the ones I described in the previous chapter, only DC- rather than AC-coupled, was given the following year by Neyer and Voges [22]. In their analysis, however, they neglect the time constants of the photodiode, amplifier, and modulator, and model the system's behavior with a difference equation. They show both numerically and experimentally that the system can exhibit steady-state, bistable, periodic, and chaotic solutions (all with frequencies less than the bandwidth of the detector), and they use an iterated map analysis to understand the conditions necessary for each type of behavior. They also note that the periodic nonlinearity of the modulator and the finite time-delay can be exploited to create a bistable and monostable multivibrator with just one optoelectronic system.

## 3.2 Bandpass Feedback

Two decades later, Larger and his collaborators began to experiment with similar types of nonlinear time-delayed chaotic generators in the interest of designing a scheme for secure chaos communication [2, 23–30]. They argued that time-delay systems are good candidates for this type of application because the chaos generated is typically very high-dimensional, which they hoped would increase the security of the message. All of their systems consist of a laser diode illuminating a nonlinear element. The resulting optical signal is incident on a photodetector, which generates an electric signal that is then amplified and fed back into the nonlinear element. As in my experiment, a MZM was the device most frequently used as the nonlinearity and the resulting oscillations can be very fast (on the order of 10 GHz). There are, however, some differences in our approaches and results: 1) they treat the electronic amplification as linear, whereas I include saturation effects; 2) they operate with feedback voltages that scan two to three fringes of the MZM transmission function [30], whereas I only have access to one transmission maximum due to the fact that  $V_{\pi,RF}$  of the modulator in my setup is on the same order as  $V_{sat}$  of the amplifier; and 3) their reported chaotic power spectra all have large peaks [27], whereas I observe a nearly “featureless” power spectrum for an operating point corresponding to the top of a transmission fringe of the MZM.

Besides the group’s advances in chaos communication, Larger and his collaborators also made two other notable contributions to the field. First, they studied both experimentally and theoretically a type of solution referred to as “breathers,” which exhibit fast-scale chaos inside a slow-scale periodic envelope [31]. These types of solutions had previously been shown to exist in ordinary differential equations (ODEs), but had yet to be found in DDEs. They argued that this type of behavior arises due to the interplay of the three very different time scales inherent to the system. I also observe

breathers in my experiments.

More recently, they modified their experimental setup with the addition of a narrow bandpass filter in the feedback loop in order to produce ultrapure microwaves [32]. This is similar to the work done by Yao and Maleki [6, 7], who used the same device to achieve stable, low-noise oscillations with frequencies as high as 75 GHz. Many applications (*i.e.*, high-speed clock recovery, comb and pulse generation, and photonic signal up and down conversion in RF systems) could benefit from oscillators that can operate in the tens of GHz regime with low noise characteristics. The principle behind the oscillator's operation is that energy is stored in the long fiber-optic delay line ( $T \sim 60$  ns) and the harmonics generated by this delay and the nonlinear modulator are suppressed by a tunable filter inserted in the feedback loop. The spectral purity of the oscillations depends on how well the oscillator can store energy, and is higher in an optoelectronic system such as this one than purely electronic systems. Electronic systems have a noise limitation due to ohmic and dispersive losses, so to compensate, one typically incorporates a high-quality-factor resonator. The resonator, however, constrains the oscillation frequency below 10 GHz. With their optoelectronic oscillator, Yao and Maleki were able to produce oscillations in the desired frequency range with a phase noise lower than -140 dBc/Hz at a 10 kHz offset, regardless of the carrier frequency.

Larger and his collaborators took the analysis one step further and showed that the amplitudes of the microwave oscillations are not always constant due to the interaction between the nonlinear element and the time delay [32], as was assumed in the previous quasi-linear analysis [6]. They found analytically (using a nonlinear dynamics approach) that, beyond a certain value of the feedback gain, the oscillations are amplitude modulated by a slowly varying envelope. This prediction was experimentally verified, where sidebands corresponding to the modulation frequency were observed

in the power spectrum. They acknowledge that an interesting problem would be to study the effect of the presence of noise in the system with a stochastic model. As discussed in Sec. 2.8, I have developed a stochastic model of the oscillator and will use the presence of noise to explain the discrepancies between my experimental observations and the predictions obtained from a noise-free model.

Illing, Blakely, and Gauthier have also analyzed nonlinear time-delayed feedback systems, both experimentally and theoretically [33–36]. The effect of bandpass filtering on the dynamics was of particular interest. Using a DDE of the form

$$\dot{x}(s) = -x(s) - y(s) + F[\gamma; x(s - \tau)], \quad (3.2)$$

$$\dot{y}(s) = \epsilon x(s), \quad (3.3)$$

where  $\epsilon$  characterizes the bandpass filter,  $\gamma$  the gain in the feedback loop, and  $\tau$  the length of the time-delay, and  $F$  is any nonlinear function of the time-delayed variable, they were able to derive a number of results valid for any system with bandpass-filtered time-delayed feedback. They showed that the single fixed point of the system is globally stable for low gain and, as the gain increases, the steady-state loses stability through a Hopf bifurcation. They showed how to calculate the Hopf curve for an arbitrary nonlinearity  $F$ , which I will later apply to the nonlinearity in my system. Also, since the gain is not uniform across the passband, periodic solutions with frequencies other than the fundamental can reach the threshold for instability first; which mode accomplishes this feat depends on the length of the time-delay. This is in contrast to the Ikeda system, which has a low-pass filter rather than a bandpass filter, where the lowest frequency mode always reach threshold first regardless of the time delay.

### 3.3 Summary

As one can see, a lot of progress has been made in the effort to understand systems with nonlinear time-delayed feedback in the past few decades. What started as a theoretical prediction of an interesting instability in such a system, has now been experimentally realized in optoelectronic systems operating at high speeds, used in a variety of applications, and further characterized theoretically. This brings us to the present, where there are still some interesting behaviors to investigate, applications to pursue, and questions to answer.

# Chapter 4

## Experimental Results

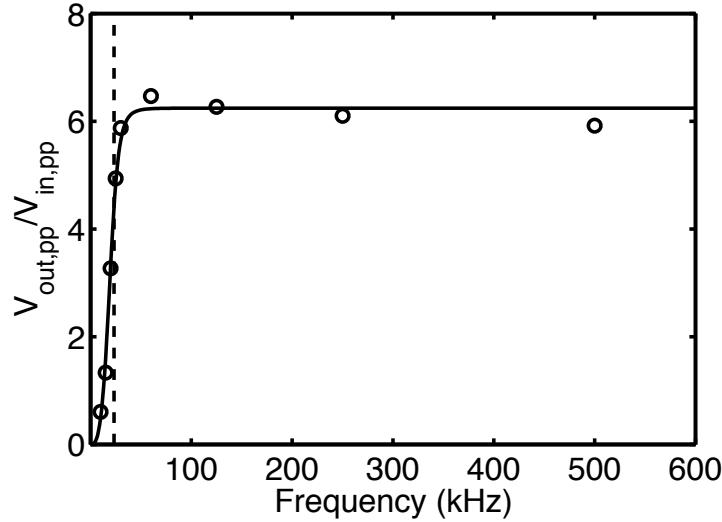
In this chapter, I describe how I measure each of the parameters in my system and report the behavior that results for different values of two of these parameters. Of particular interest are the dynamics that occur for  $m = 0$ , where the system transitions from a linearly stable steady-state solution into broadband chaotic behavior.

### 4.1 Experimental Parameters

As previously discussed, the dynamics of the physical system under investigation depend on magnitude of five dimensionless quantities related to: the frequency characteristics of the filter, the saturation characteristics of the modulator driver, the length of the time delay, the feedback strength, and the operating point of the nonlinearity. The values of the first three quantities mentioned are fixed in most of my experiments, although the length of the time delay is occasionally altered when extra elements are added to the feedback loop. I discuss the determination of each of these five parameters in the following subsections.

#### 4.1.1 Fixed Parameters

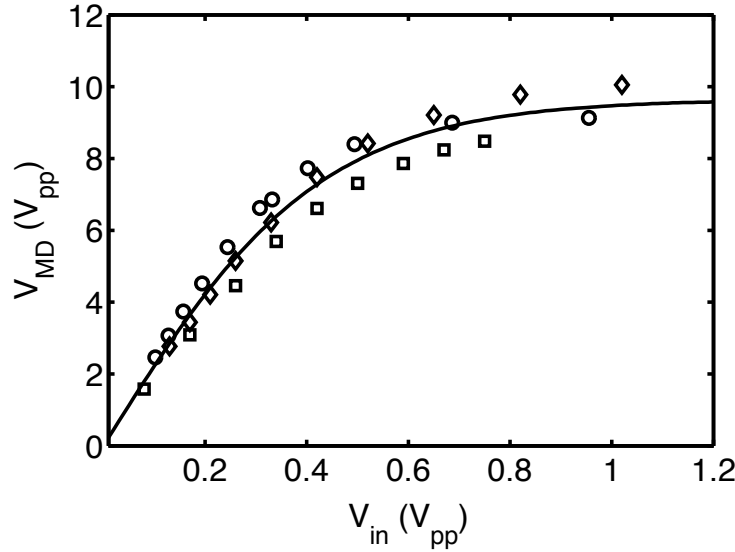
As Eq. 2.57 indicates,  $\epsilon$  depends on the system's bandpass characteristics. I approximate the bandwidth of the system by determining the high- and low-pass characteristics for signals propagating through the entire feedback loop. By opening up the feedback loop and injecting sinusoidal signals with frequencies ranging from 10-500



**Figure 4.1:** The frequency-dependent gain of the feedback loop with a third-order high-pass filter fit superimposed. The corner frequency is 23.4 kHz, as denoted with the dashed vertical line, and the gain of the open-loop system is 6.24.

kHz into the electronic splitter, I record the frequency dependent gain. With this data, Lucas Illing determined the 3 dB high-pass corner frequency to be 23.4 kHz by fitting the data points third-order high-pass filter, as shown in Fig. 4.1. I was not able to produce signals in the tens of GHz range, so I have to determine the low-pass corner frequency via a different method. By looking at the spectrum of the system's output in a high-speed spectrum analyzer with a bandwidth of 20 GHz borrowed from another laboratory, I can deduce information about the frequency response. (Unfortunately, I was not able to reproduce the spectra obtained with this device here because the spectrum analyzer could not output the recorded waveform.) The spectrum of the broadband chaos I will later report on has a distinct rolloff at 12 GHz. Since the measuring device can detect frequencies beyond this range, it must be the system itself that is limiting the bandwidth of the output. I therefore take 12 GHz to be a rough estimate for the low-pass corner frequency of the filter in my model. Taken together with the

high-pass frequency, the frequency of maximum transmission and the bandwidth of the filter are determined. These two quantities define  $\epsilon$  according to Eq. 2.57, which I find to be  $1.9 \times 10^{-6}$ . Note that both the high- and low-pass frequencies determined for the feedback loop are inconsistent with the high- and low-pass frequencies quoted by the manufacturer for the modulator driver (30 kHz and 10 GHz, respectively). As I did not measure the frequency characteristics of the modulator driver individually, it is likely that the particular driver I use has high- and low-pass frequencies close to what I report here for the effective filter of the overall system.



**Figure 4.2:** The amplitude-dependent gain of the modulator driver measured at three frequencies: 240 MHz (squares), 1 GHz (diamonds), and 2 GHz (circles).

I determine the linear gain  $g_{MD}$  and saturation voltage  $V_{sat}$  of the modulator driver by injecting 140 MHz, 1 GHz, and 2 GHz sinusoidal signals into its input and recording the amplitude dependent gain (*i.e.*, the ratio of the peak-to-peak output voltage to the peak-to-peak input voltage). The frequencies and amplitudes of the signals were measured using a digital oscilloscope (Agilent 54853A) with a bandwidth of 2.5 GHz.

For low input amplitudes, the data points fall approximately on a line, as shown in Fig. 4.2, but for high input amplitudes the gain saturates. Lucas Illing fit the data to a hyperbolic tangent (Eq. 2.53) using a Fletcher-Powell curve fitting method, which yielded a value of 9.65 V for  $V_{sat}$  and a value of 22.6 for  $g_{MD}$ .  $V_{sat}$  will eventually be used to determine the dimensionless parameter  $d$ , once  $V_{\pi,RF}$  of the MZM is known, while  $g_{MD}$  will help determine  $\gamma$ .

Recall that  $V_{\pi,RF}$  characterizes the width of an interference fringe produced by the RF port of the MZM. To determine this voltage, I set the laser to output a constant power, bias the MZM at the bottom of a fringe, and apply a few nanosecond pulse to the RF port with a function generator (Tektronix AFG3251). After receiving the optical signal with the Miteq photodetector and measuring the resulting voltage with the 8 GHz digital oscilloscope, I determine the input pulse amplitude that generates the maximum output pulse amplitude. This amplitude is equal to  $V_{\pi,RF}$ , which I find to be 7.40 V. Taken together with  $V_{sat}$ , Eq. 2.60 gives  $d = 2.05$ .

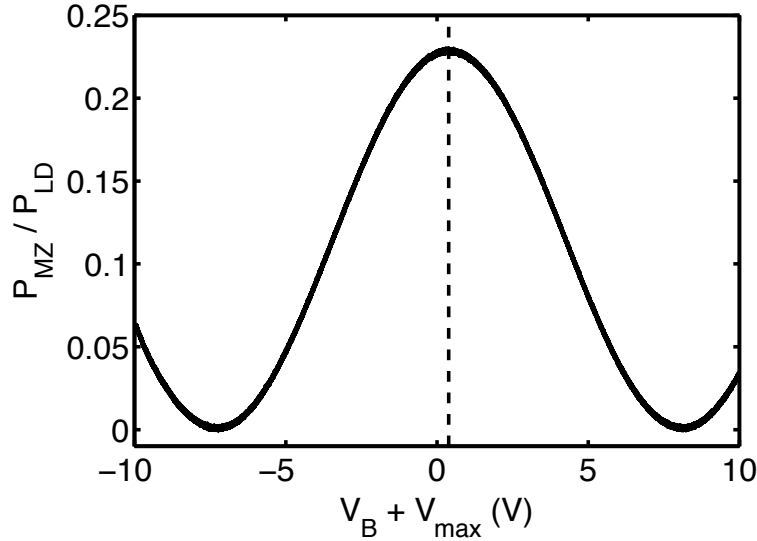
To measure the time delay, I once again opened up the feedback loop and injected a pulse with a full-width at half max (FWHM) of 5 ns into the electronic splitter. The two resulting pulses reached their final destination (the 8 GHz oscilloscope) via two different paths, with one having to travel the additional distance of the feedback loop. By looking at their time separation and taking into account any length differences in the cables used to send the signals to the oscilloscope, I determined the time delay to be  $24.0 \pm 0.3$  ns, where the error is approximated by the standard deviation of the time delay measurements. This measurement, however, neglected one small connector that is typically incorporated in the feedback loop. In later closed-loop measurements, I see features that indicate a time delay of 24.1 ns. Since this measurement incorporates the complete feedback loop and the value falls well within the statistical experimental uncertainty of the original measurement, I take it to be the length of the time delay.

Then, according to Eq. 2.61,  $\tau = 1820$ .

### 4.1.2 Free Parameters

The last two quantities (the operating point and feedback strength) are easy to measure and adjust. One could quantify these in terms of physical parameters, such as the bias voltage of the MZM and power of the laser, but, in order to easily compare experiment and theory, I again report their dimensionless counterparts.

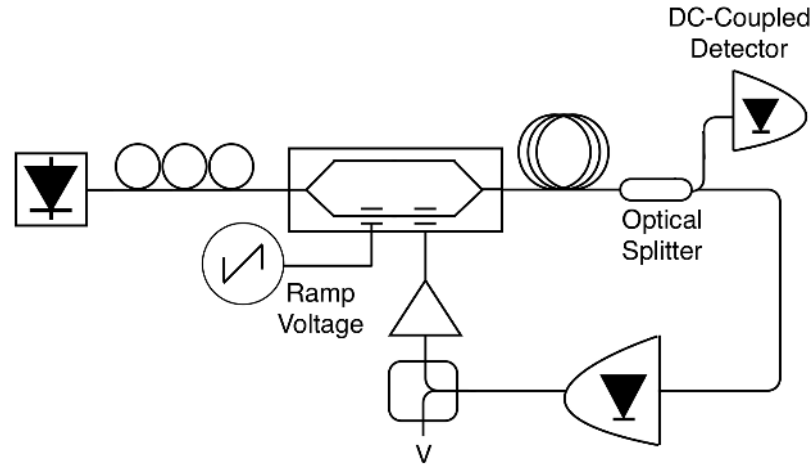
According to Eq. 2.59,  $m$  is proportional to  $V_B$  and inversely proportional to  $V_{\pi,DC}$ . Recall that  $V_B$  is related to  $V_{max}$  and the voltage applied to the DC port of the MZM with a digital power supply, while  $V_{\pi,DC}$  characterizes the width of an interference fringe produces by the DC port of the MZM.



**Figure 4.3:** The fractional power transmitted through the MZM as a function of the applied voltage. The dashed vertical line intersects the horizontal axis at  $V_{max}$ .

The applied voltage  $V_{max}$  that produces a transmission maximum is determined experimentally by applying a ramp voltage to the bias port that changes linearly from

-10 V to 10 V and recording the output on the 8 GHz digital oscilloscope. If the period of the applied signal is known, one can use the oscilloscope to measure  $V_{max}$ , as shown in Fig. 4.3. Unfortunately, not only is  $V_{max}$  different for each MZM, but, for a particular MZM, it can drift by as much as a Volt with changes in temperature induced by the signal driving the RF port of the MZM. Thus,  $V_{max}$  needs to be measured before  $V_B$  can be set for each experimental run.

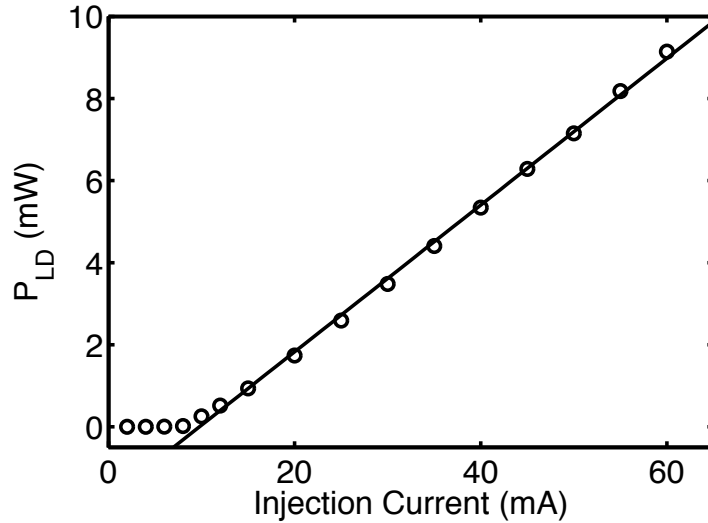


**Figure 4.4:** Schematic depicting how  $V_{max}$  can be determined by applying a ramp voltage to the DC port.

To allow for a quick measurement of  $V_{max}$ , I alter the setup with the inclusion of an optical splitter, as shown in Fig. 4.4. A small portion of the optical signal exiting the MZM is sent to a DC-coupled photodetector (New Focus 2011) and the rest propagates through the feedback loop. By applying a slowly-varying ramp voltage to the DC port with a function generator (Stanford Research Systems DS355) and keeping the overall system gain relatively low so that the system remains in steady state, I am able to reproduce the expected transmission curve (shown in Fig. 4.3) and determine  $V_{max}$ . For a sufficiently slow frequency of the ramp voltage ( $\sim 100$  Hz), I can also determine  $V_{\pi,DC}$  from the time separation between  $V_{max}$  and the first interference minima. With

this method, I find that  $V_{\pi,DC}$  is 7.68 V.

As Eq. 2.58 indicates, the dimensionless feedback gain  $\gamma$  depends on the power incident on the MZM and the transmission characteristics of each device in the feedback loop ( $g_{MZ}$ ,  $g_{S0}$ ,  $g_{PD}$ ,  $g_{S1}$ , and  $g_{MD}$ ). For the setup depicted in Fig. 2.1, the power incident on the MZM is simply the power from the laser diode (assuming no insertion loss in the polarization controller), which can be determined from the injection current. To determine this dependence, I measured the output power with a photoreceiver (Thorlabs DET01CFC) for several values of the current, as shown in Fig. 4.5. This photoreceiver outputs a voltage, which I measure with the 8 GHz digital oscilloscope, and then convert to a power by dividing by a factor of  $R = 47.5$  V/W, where  $R$  is the optical to electronic gain given by the manufacturer. I find that, above  $\sim 10$  mA, the measured steady-state power (in milliwatts) as a function of current (in milliamperes) for my laser is approximately given by

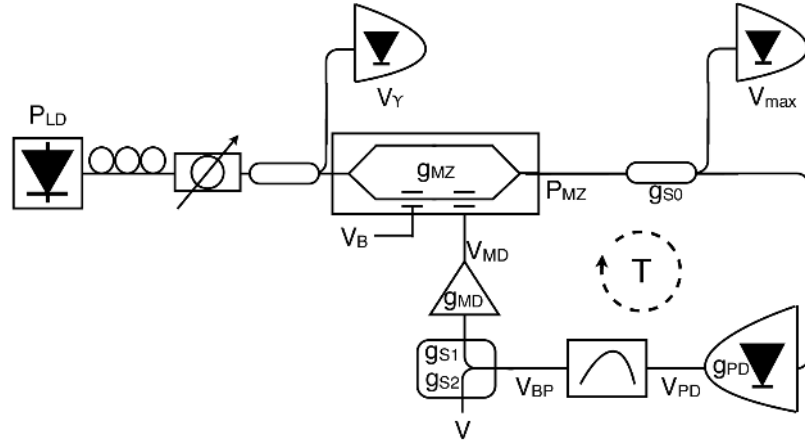


**Figure 4.5:** Dependence of laser power on injection current with a least-squares-fit in the linear regime superimposed.

$$P_{LD} \approx 0.179 \text{ [mW/mA]}(I - I_{th}), \quad (4.1)$$

with  $I_{th} = 9.77 \text{ mA}$ , based on a least-squares fit for the data points in the linear regime. The excellent visual agreement validates the approximations made in Sec. 2.1.2 for the steady-state behavior of the laser diode.

Alternatively, I can split the signal after the polarization controller with a 50/50 optical splitter, and measure the power incident on the MZM directly using the Thorlabs photoreceiver in order to determine  $\gamma$ , as shown in Fig. 4.6. For the experiments described in the following sections, this is the method I use.

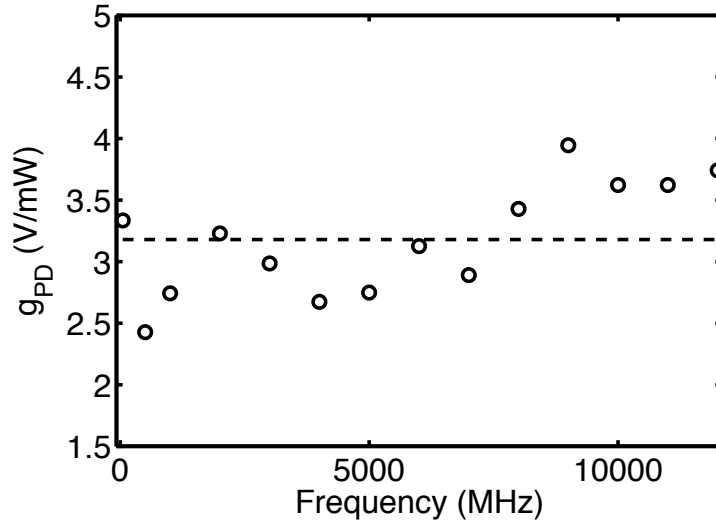


**Figure 4.6:** Schematic of setup used in the experiments with the characteristics of each component labeled. The addition of the variable attenuator and two optical splitters allow the user to quickly determine  $V_\gamma$  and  $V_{max}$ .

To determine the insertion loss of the MZM  $g_{MZ}$ , I need to find the ratio of the maximum power out of the MZM to the power incident on the MZM. I measure these optical powers with the Thorlabs photoreceiver, and  $g_{MZ} = 0.23$  when the incident light is properly polarized. The insertion loss of the optical splitter  $g_{S0}$  is simply the ratio of output power to input power. I also measured this quantity with the Thorlabs

photoreceiver and the 8 GHz digital oscilloscope, and fin  $g_{s0} = 0.67$ .

The optical-to-electrical transfer characteristics of the specific photodetector used in my experiments are provided by the manufacturer over a range of frequencies, as shown in Fig. 4.7. Since the dynamics of the oscillator can take on any of these frequencies, I compute the average, giving  $g_{PD} = -3.2$  V/mW, where the negative sign indicates that the detector is inverting. The insertion losses of the two arms of the power splitter are also given by the manufacturer, however, not for the particular component used in the setup. The manufacturer claims that, for each device, the insertion loss is within 0.5 dB of the 6 dB average loss. I, therefore, take  $g_{s1}$  and  $g_{s2}$  to be 0.50.



**Figure 4.7:** The optical-to-electrical transfer characteristics for the Miteq photodetector as quoted by the manufacturer. The dashed line indicates the average, which I take to be  $g_{PD}$ .

I now have enough information to determine  $\gamma$  based on the voltage  $V_\gamma$  produced

by the Thorlabs receiver through the relation

$$\gamma = \frac{g_{MZ}g_{PD}g_{S0}g_{S1}g_{MD}}{RV_{sat}}V_{\gamma}, \quad (4.2)$$

$$= 0.0126[\text{mV}^{-1}]V_{\gamma}. \quad (4.3)$$

### 4.1.3 Parameter Error Estimation

The parameters previously described are subject to numerous sources of error. The first fixed parameter,  $\epsilon$ , is a function of the high- and low-pass frequencies of the filter,  $f_-$  and  $f_+$ . The low-pass frequency could only be roughly determined, so I estimate the uncertainty in  $f_+$  to be 0.5 GHz. While the uncertainties from both  $f_+$  and  $f_-$  contribute to the error in  $\Delta$ ,  $\omega_0$ , and  $\epsilon$ , the uncertainty in  $f_+$  is so large that it will dominate for any reasonable estimate in the uncertainty of  $f_-$ . I thus find the error in  $\Delta$  to be about 4%, the error in  $\omega_0$  to be about 2%, and the error in  $\epsilon$  to be about 9%.

The second fixed parameter,  $d$ , is a function of  $V_{sat}$  and  $V_{\pi,RF}$ . I estimate the error in  $V_{sat}$  based on the quality of the fit shown in Fig. 4.2 and I find it to be approximately 0.5 V. Based on the experiment I perform to determine  $V_{\pi,RF}$ , I estimate its uncertainty to be 50 mV. These errors combine to give an uncertainty in  $d$  of about 5%.

The dimensionless time  $\tau$  is a function of  $T$  and  $\Delta$ , where the uncertainty in  $\Delta$  is discussed above. The error in  $T$  is estimated experimentally to be 0.3 ns. Combining these errors, gives  $\tau = 1820 \pm 80$ .

The value of  $m$  changes from experiment to experiment and depends on  $V_B$  and  $V_{\pi,DC}$ . I determine  $V_{max}$  for each experiment, as described in the previous subsection, with an estimated accuracy of 20 mV. I therefore take 20 mV to be my error in  $V_B$ .  $V_{\pi,DC}$  is determined in a similar fashion, and I estimate the error in this quantity to also be 20 mV. The total error in  $m$  will depend on these errors as well as the magnitude of  $V_B$ . For

the range of values used in the experiments, the error in  $m$  ranges from approximately 0.004 to 0.006.

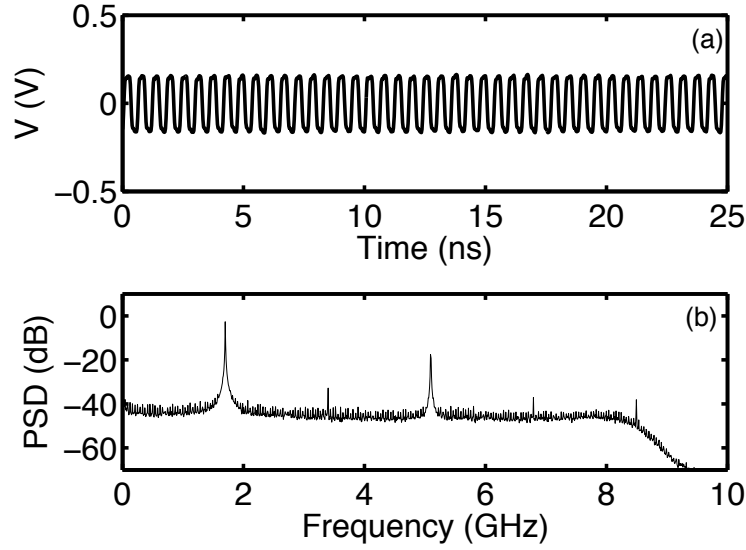
The dimensionless feedback gain  $\gamma$  is a function of several quantities, with one of these quantities ( $V_\gamma$ ) free to vary. The fixed quantities are  $R$ ,  $g_{MZ}$ ,  $g_{S0}$ ,  $g_{PD}$ ,  $g_{S1}$ ,  $g_{MD}$ , and  $V_{sat}$ . Here, I compute the percent uncertainty in  $\gamma$  due to the estimated error for each of the fixed quantities, and later in the chapter I will calculate the contribution from the error in  $V_\gamma$ . I will assume that the Thorlabs detector is functioning according to the manufacturers specifications, so that there is no uncertainty in  $R$ . I estimate the uncertainties in  $g_{MZ}$  and  $g_{S0}$  to be 0.01, as the percent transmission through each of these devices can change slightly when fibers are connected or disconnected. To find the uncertainty in  $g_{PD}$ , I take the standard deviation of the values shown in Fig. 4.7, which is 0.46 V/mW. The uncertainty in  $g_{S1}$  is quoted to be 0.03 by the manufacturer. Finally, I again use the quality of the fit for the modulator driver saturation, and estimate the uncertainty in  $g_{MD}$  to be 1. Taken together with the error in  $V_{sat}$  discussed earlier, I find the total relative error in  $\gamma$  to be approximately 18%.

## 4.2 Observed Dynamics

I see a wide range of behaviors as I vary the two free parameters of my system. In addition to the steady-state behavior dominant at low feedback gain, I also see periodic oscillations, quasi-periodic oscillations, breathers, chaos, and broadband chaos. Many of these behaviors have already been discovered and occur for values of  $|m|$  in the range of  $\pi/16$  to  $7\pi/16$ . I refer to these as the traditional operating points. The broadband chaos, which, to the best of my knowledge, has yet to be reported in the literature, occurs for  $m$  near 0 and  $\pi/2$ . I refer to these as the new operating points. For all of my experiments, I measure  $V_{max}$  as described in the previous section and then

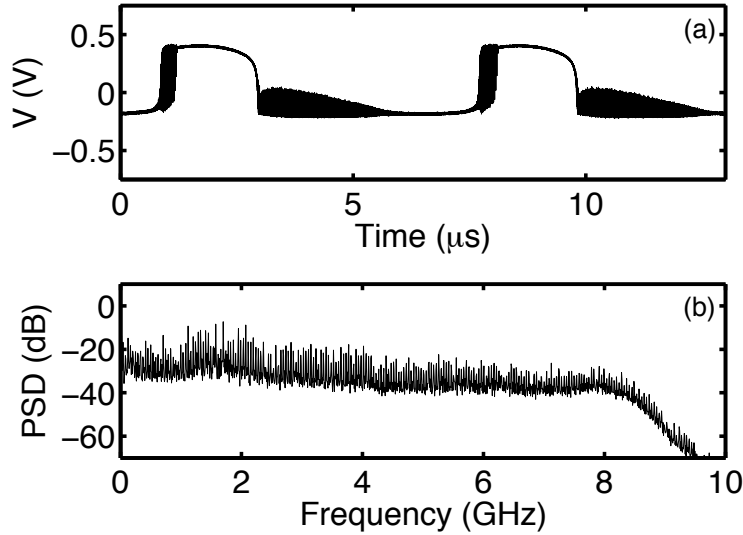
set  $V_B$  with a power supply while the system is in steady-state. I then vary  $\gamma$  using the variable optical attenuator and monitor its value according to the output voltage of the Thorlabs detector.

#### 4.2.1 Traditional Operating Points



**Figure 4.8:** Periodic solution obtained from the experimental system with parameter values  $m = -\pi/4$  and  $\gamma = 0.94$ .

Examples of experimental time series for different types of behavior at the traditional operating points are shown in Figs. 4.8 and 4.9. The time series in this subsection and the following subsection are recorded with the 8 GHz digital oscilloscope at a sampling rate of 40 GSa/s. The one-sided power spectral densities (PSDs) are computed by taking a fast Fourier transform (FFT) of the time series, then normalized such that the mean-squared amplitude power of the signal in the time domain is equal to that in the frequency domain, and finally subjected to a median filter with a filter bin width of about 8 MHz.

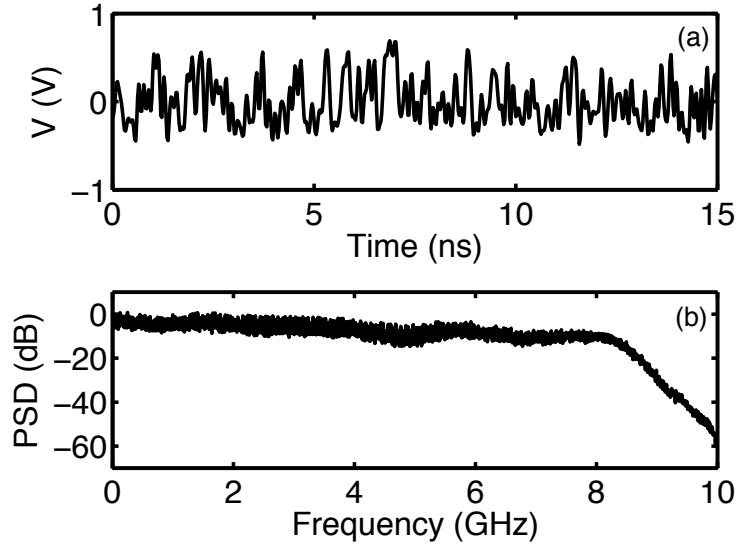


**Figure 4.9:** Breather solution obtained from the experimental system with parameter values  $m = -0.23$  and  $\gamma = 1.3$ .

### 4.2.2 New Operating Points

For sufficiently high feedback gain at  $m \approx 0$  and  $m \approx \pi/2$ , the system exhibits high-speed and broadband chaotic dynamics. Time series and power spectra at both of these operating points are shown in Figs. 4.10 and 4.11. For the rest of the discussion, however, I will focus solely on the  $m = 0$  operating point, keeping in mind that similar dynamics could occur for  $m = \pi/2$ .

The important thing to note is that the power spectrum at this operating point is “featureless,” as it is essentially flat up to the cutoff frequency of the oscilloscope used to measure the dynamics (8 GHz), indicating that all frequencies are contributing with approximately the same power. This should be compared to the power spectrum of the (noisy) steady-state behavior just below the transition point shown in Fig. 4.12. Over the 8 GHz frequency range, the broadband chaos spectrum is contained in a 15 dB range with a standard deviation of 3 dB, while the spectrum of the noise falls

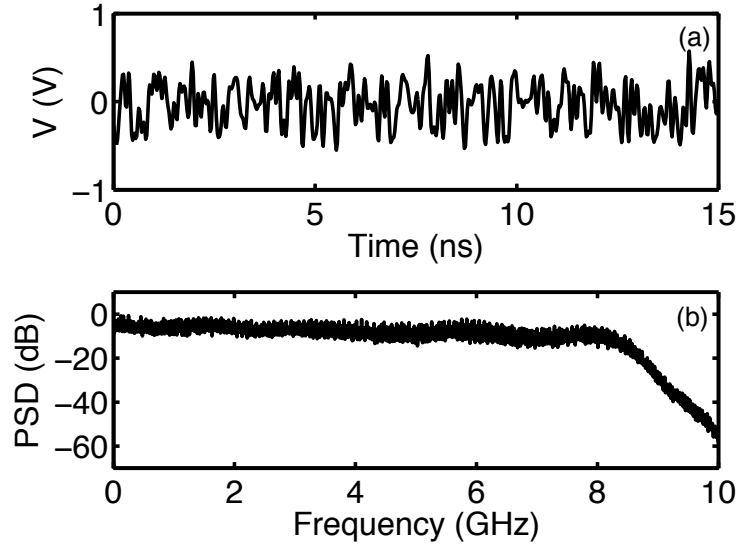


**Figure 4.10:** Time series (a) and power spectrum (b) for broadband chaos in the physical system with parameter values of  $m = 0$  and  $\gamma = 4.80$ .

within a 18 dB range with a standard deviation of 2 dB, indicating that the degrees of “flatness” are similar. The chaotic fluctuations, however, have a much larger peak-to-peak amplitude than the noise fluctuations.

Typically, the power spectrum of a chaotic time series will have broad peaks corresponding to unstable orbits, giving one a sense of the structure of the attractor. It is still uncertain what the broadband power spectrum shown here indicates. It could be that there are a large, or even infinite, number of unstable periodic orbits that make up the attractor.

Another interesting feature of this new operating point is that the steady-state solution of the noise-free model is linearly stable for all values of the feedback gain, as I will show in Sec. 4.1. Experimentally, however, I find that the steady-state behavior eventually transitions into broadband chaotic behavior as I increase the gain beyond a certain threshold. Thus, I hypothesize that the presence of noise lowers a bifurcation



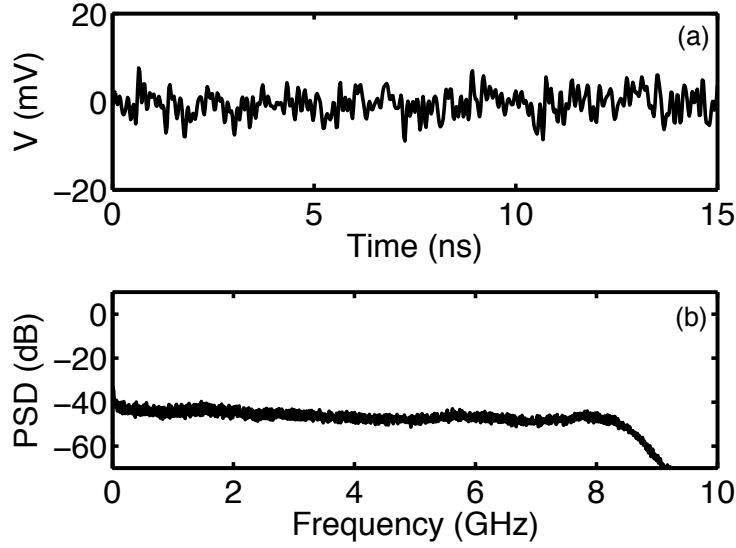
**Figure 4.11:** Time series (a) and power spectrum (b) for broadband chaos in the physical system with parameter values of  $m = \pi/2$  and  $\gamma = 4.23$

threshold from infinity to a finite and experimentally accessible value.

### 4.2.3 Gain Threshold

To quantify the phenomenon discussed in the previous subsection, I measure the values of the gain for which the physical system transitions away from or back to steady-state behavior as a function of  $m$ . The easiest way to tune the gain is by varying the injection current of the laser diode. As discussed in Sec. 2.1.3, however, the noise characteristics change dramatically with the current, which is why I insert a variable optical attenuator between the polarization controller and the MZM and use this to adjust the gain while keeping the injection current constant at 150 mA.

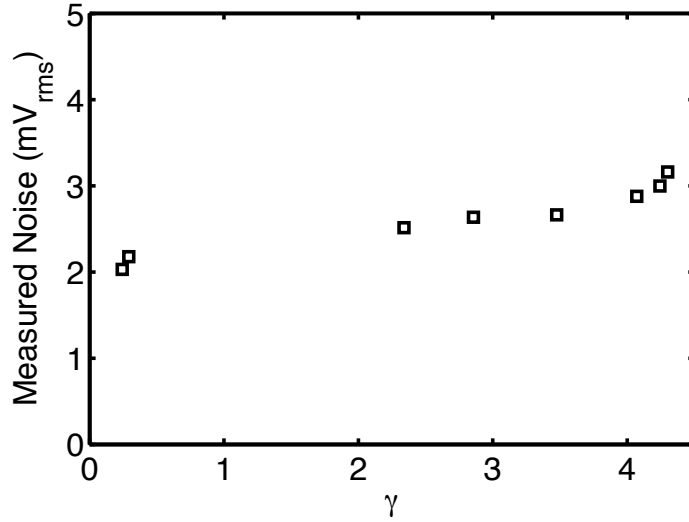
The noise from the laser diode and detector are also affected by the attenuation, but the effect is less dramatic for the values of  $\gamma$  where bifurcations occur. While in steady-state at  $m = 0$ , I measured the root-mean-square of the voltage fluctuations for



**Figure 4.12:** Time series (a) and power spectrum (b) for the noisy steady-state of the physical system with parameter values of  $m = 0$  and  $\gamma = 4.30$ . Note the difference in scale between (a) and Figs. 4.10(a) and 4.11(a).

values of  $\gamma$  between 0.24 and 4.3 using the 8 GHz digital oscilloscope. As can be seen in Fig. 4.13, the noise level ranged from approximately  $2\text{-}3 \text{ mV}_{\text{rms}}$ , indicating that the noise amplitude changes very little over the region of interest.

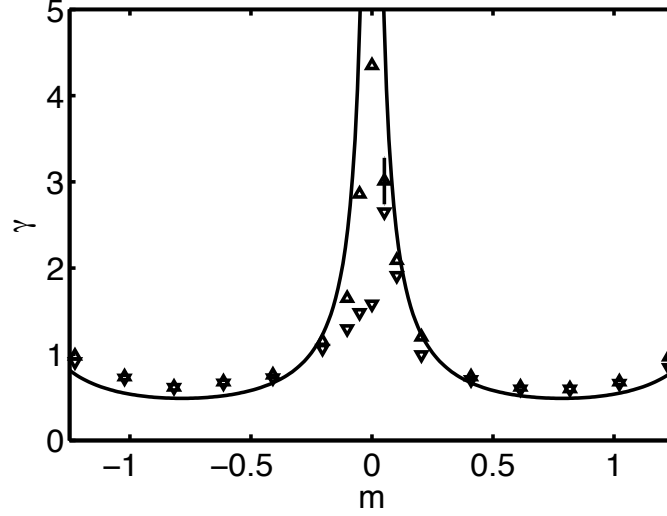
Because the bifurcations away from steady-state can either be supercritical or subcritical, hysteresis can occur. There are therefore two quantities of interest for this experiment: the gain for which the steady-state loses stability as it is increased ( $\gamma_{th+}$ ) and the gain for which the steady-state regains stability as it is decreased ( $\gamma_{th-}$ ). I measure both quantities as a function of  $m$  using the 8 GHz digital oscilloscope by recording the value of  $\gamma$  where I see a bifurcation occur for each  $m$ . I repeat this measurement five times. When the bifurcation is subcritical, the amplitude of the oscillatory solution changes abruptly and it is therefore easy to determine  $\gamma_{th}$ . For supercritical bifurcations, however, the amplitude of the oscillatory solution grows gradually from zero, making the identification of the transition prone to some systematic error. To make



**Figure 4.13:** Experimental noise measured with the 8 GHz oscilloscope at  $m = 0$  as a function of  $\gamma$ .

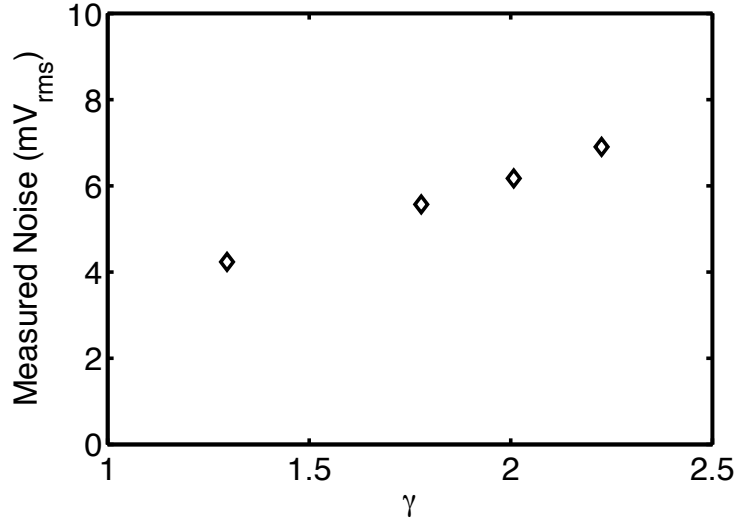
my measurements more consistent, I use the noise floor to set the transition threshold: when the oscillations grow above the noise floor and the root-mean-square voltage recorded by the oscilloscope changes, a transition has occurred. Figure 4.14 shows that, near  $m = 0$ , where the bifurcation are subcritical, the system is very hysteretic, as there is a significant overlap between stable steady-state and oscillatory behavior. For values of  $m$  near  $\pi/4$ , the bifurcations are supercritical and there is practically no overlap.

It is also interesting to note that the maximum values of  $\gamma_{th+}$  and  $\gamma_{th-}$  are obtained for  $m$  slightly greater than zero. The asymmetry in  $\gamma_{th-}$  is much more pronounced and can be seen in Fig. 4.14, while the asymmetry in  $\gamma_{th+}$  is only evident when I measure it over a much finer scale, as shown in Fig. 4.16. I will later show that, in this particular case, one expects the bifurcation curve obtained from the noise-free to be symmetric about  $m = 0$ . While it is not yet understood why the maximum for  $\gamma_{th+}$  is shifted



**Figure 4.14:** Experimentally determined  $\gamma_{th+}$  (triangles) and  $\gamma_{th-}$  (upside-down triangles) as a function of  $m$ . The bifurcation curve derived from the noise-free model,  $\gamma_H$ , is drawn with a line. The relative uncertainty in  $\gamma$  is approximated with the standard deviation. The average percent uncertainty is 2%, and the maximum is 9%. Only the maximum is shown with an error bar, as most error bars would not extend beyond the data marker.

towards positive  $m$ , I will provide some evidence for why one would expect  $\gamma_{th-}$  to be asymmetric. To explore the effect of noise further, I measure the value of  $\gamma_{th+}$  near  $m = 0$  for two different noise strengths. The noise in the lower-noise experiment was due to the previously discussed noise sources and ranged from about 2-3 mV<sub>rms</sub>. To increase the noise for the second experiment, I inserted an Erbium doped fiber amplifier (EDFA) before the MZM (so as not to change the length of the time delay). Like their name indicates, EDFAs are typically used to amplify an optical signal, but will also add noise as a by-product. It is the additional noise, and not the amplification, that make it suitable for the purposes of my experiment. I placed a variable attenuator in the setup (after the EDFA and before the polarization controller) and used this to control gain, just like in the low-noise experiment. This time, however, the root-mean-

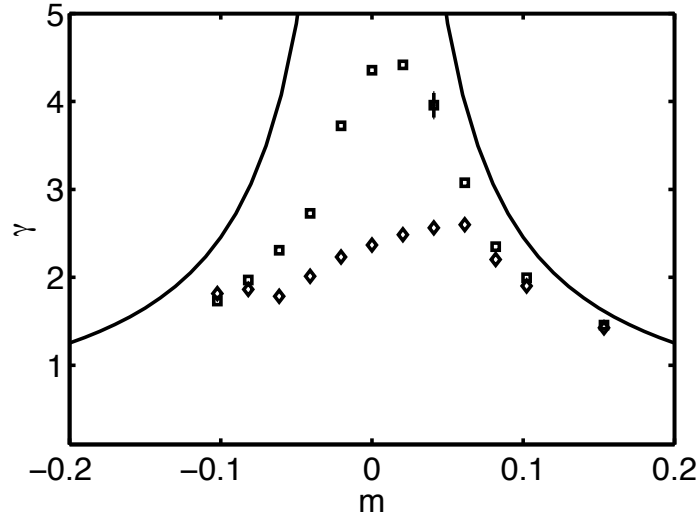


**Figure 4.15:** Experimental noise measured with the 8 GHz oscilloscope at  $m = 0$  as a function of  $\gamma$ . The addition of an EDFA increases the noise level by a factor of about two from the previous experiment.

square of the voltage fluctuations in the region of interest were approximately twice as large, ranging from about 4-7  $\text{mV}_{\text{rms}}$ , as shown in Fig. 4.15. As expected, the threshold for increasing gain is lower for the experiment with higher noise than in the experiment with lower noise, as shown in Fig. 4.16.

#### 4.2.4 Transient Behavior

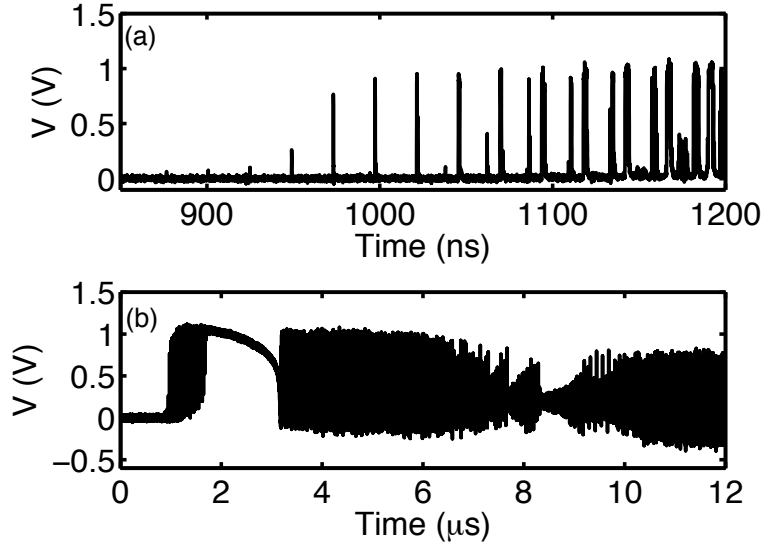
In the interest of understanding the transition from the steady-state behavior at  $m = 0$ , I record the transient behavior that takes place as the gain is increased beyond  $\gamma_{th+}$ . As one can see from Fig. 4.17, the initial transient takes the form of a series of narrow pulses separated in time by  $T$  and with a FWHM of  $\sim 0.2$  ns. These pulses grow in amplitude initially, but the amplitude remains approximately constant after about the fourth pulse. Around this time, a second train of pulses appears to emerge, also with a



**Figure 4.16:** Experimentally determined  $\gamma_{th+}$  for a low (squares) and high (diamonds) noise levels, with  $\gamma_H$  (line) superimposed. The relative uncertainty in  $\gamma$  is approximated with the standard deviation. The average percent uncertainty is 1.5%, and the maximum is 4%. Only the maximum is shown with an error bar, as most error bars would not extend beyond the data marker.

time separation of  $T$ .

To quantify more precisely how a finite perturbation drives the system away from the linearly-stable steady-state, I manually inject pulses with a FWHM of 0.2 ns into the feedback loop while the system is in steady-state (*i.e.*, for  $\gamma < \gamma_{th+}$ ). Using a high speed pulse generator and an additional power splitter, I combine pulses of various amplitudes with the signal exiting the photodetector, as shown in Fig. 4.19. In general, I find that an input pulse with sufficient amplitude generates a train of pulses separated in time by  $T'$ , where  $T'$  is the length of the time delay with the additional tide delay caused by the extra power splitter, as shown in Fig. 4.18. Depending on  $\gamma$  and the amplitude of the initial pulse, the subsequent pulse train will either decay back to the steady-state (Fig. 4.19(a)) or grow in amplitude (Fig. 4.19(b)). I record the amplitudes

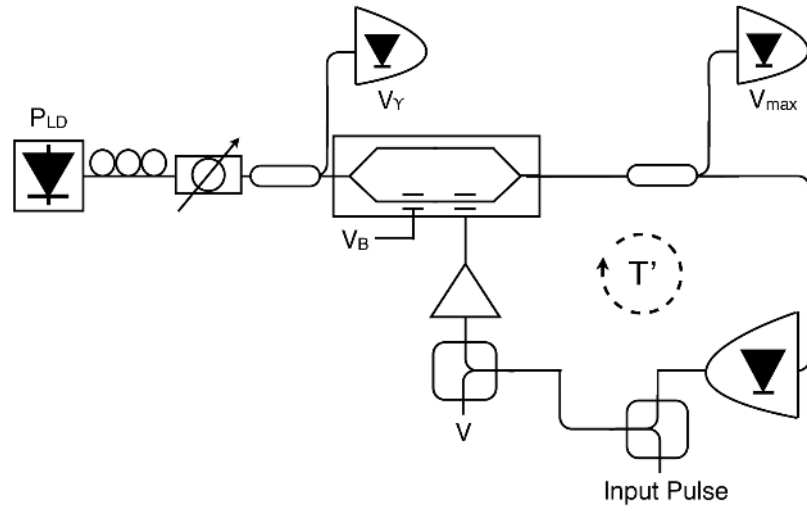


**Figure 4.17:** The transient behavior that occurs in the physical system when the fixed point first loses stability for parameter values  $m = 0$  and  $\gamma = 4.36$ . The initial pulse train highlighted in (a) leads into the breather-like behavior shown in (b).

$V_{th,exp}$  at which this transition occurs five times for each value of  $\gamma$ , and the results are shown in Fig. 4.20. The features of these pulse trains will be exploited in the next chapter to understand how the presence of a small amount of noise can alter what would otherwise be steady-state behavior.

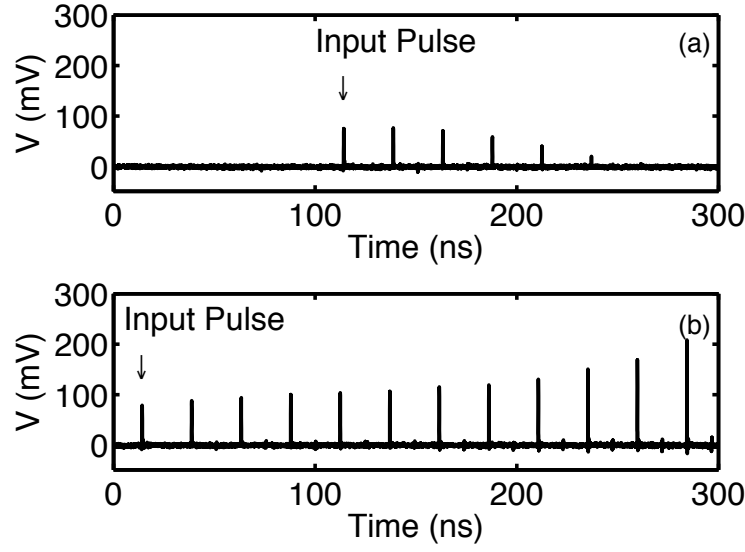
### 4.3 Summary

In this chapter, I show that my physical system, which is subject to experimental noise, displays some interesting behavior for an operating point of  $m = 0$ . Namely, the frequency spectrum of the dynamics at this point is featureless and the threshold for obtaining such behavior lowers as the noise in the system is increased. Furthermore, the initial transition from steady-state to the broadband chaotic behavior takes the form of a train of narrow pulses with growing amplitude separated in time by  $T$ . This behavior

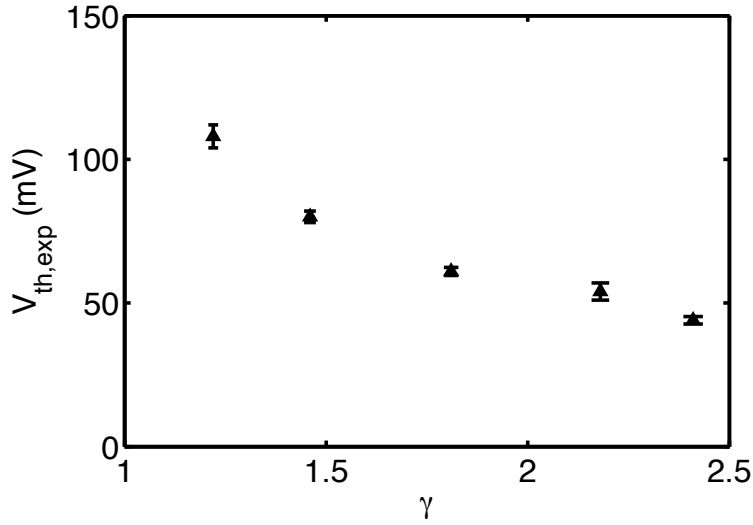


**Figure 4.18:** Schematic of setup used to inject pulses into the feedback loop.

is replicated by injecting pulses into the feedback loop and the critical amplitude for which steady-state behavior is lost is determined for several values of  $\gamma$ .



**Figure 4.19:** Two pulse trains generated by injecting pulses with amplitudes of 75.1 mV (a) and 78.7 mV (b) into the feedback loop of the physical system with the additional power splitter.



**Figure 4.20:** Experimentally determined  $V_{th,exp}$ . The error bars indicate the uncertainty, which is approximated with the standard deviation of the measured values. The relative uncertainty in  $\gamma$  is less than one percent and therefore is not shown. The original system under the influence of noise goes unstable for  $\gamma = 4.36$ .

## Chapter 5

### Analysis of the Model

To gain further insight into the system dynamics, I start by studying the stability of the single fixed point of Eqs. 2.64 and 2.65 using linear stability analysis. I find that the fixed point of this noise-free model is stable for zero gain and will ultimately lose stability via a Hopf bifurcation as the gain is increased beyond some threshold, with the exception of one particular operating point. I then report the results obtained by numerically integrating the physical model using a four-point Adams-Bashforth-Moulton scheme, which was found to produce solutions similar to those observed experimentally. In an effort to explain the origin of these dynamics, I study the phase space of the dimensionless model. The results from this analysis, as well as those obtained from experiments, motivate the creation of a one-dimensional map that provides rough criteria for when the dynamics at  $m = 0$  change dramatically due to the influence of noise.

Several people contributed to the analysis that follows. Illing and Gauthier's work on bandpass-filtered feedback with a general nonlinearity [34] provided the groundwork for generating the Hopf curve, which I apply to the specific nonlinearity in my system. Zheng Gao supplied me with code to numerically integrate a model incorporating bandpass-filtered feedback. Eckehard Schoell provided much insight to the phase space analysis of the dimensionless model, where he used these results to reduce the DDE to a one-dimensional map that I analyze.

## 5.1 Linear Stability Analysis

The single fixed point  $(x^*, y^*)$  of Eqs. 2.64 and 2.65 is found by setting the derivatives equal to zero, giving  $x^* = 0$  and  $y^* = 0$ . The linear stability of the steady-state can then be determined by Taylor-series expanding the nonlinearity about this fixed point, assuming a small perturbation of the form  $\delta y = e^{\lambda s}$ , and analyzing the resulting characteristic equation

$$\lambda^2 + \lambda + \epsilon + b\lambda e^{-\lambda\tau} = 0, \quad (5.1)$$

where  $b = -\gamma d \sin(2m)$  is the effective slope of the nonlinearity in the vicinity of the fixed point. Here,  $\lambda$  represents the infinite number of eigenvalues that characterize the stability of the system. The fixed point loses stability when  $\Re[\lambda]$  becomes positive. Thus, by setting  $\Re[\lambda] = 0$  and  $\Im[\lambda] = i\Omega$ , I can determine the stability of the fixed point from

$$(i\Omega)^2 + i\Omega + \epsilon + b(i\Omega)e^{-i\Omega\tau} = 0. \quad (5.2)$$

Separating the real terms from the imaginary terms gives the following set of stability equations

$$-\Omega^2 + \epsilon - b\Omega \sin(\Omega\tau) = 0, \quad (5.3)$$

$$1 - b \cos(\Omega\tau) = 0. \quad (5.4)$$

Note that these equations remain unchanged for  $\Omega \rightarrow -\Omega$  and, for  $\epsilon > 0$ , there is no solution for  $\Omega = 0$ . This implies that the eigenvalues cross the imaginary axis in complex conjugate pairs and, thus, the steady-state loses stability via a Hopf bifurcation.

As  $\tau$  and  $\epsilon$  are fixed for a given experimental setup, the stability equations can be used to solve for the values of  $\Omega$  and  $b$ , where the fixed point of the noise-free model undergoes a Hopf bifurcation.

To determine  $\Omega$  as a function of  $b$ , I solve Eq. 5.3 for  $\sin(\Omega\tau)$  and Eq. 5.4 for  $\cos(\Omega\tau)$ , and add the squares of these results to obtain

$$1 = \sin^2(\Omega\tau) + \cos^2(\Omega\tau), \quad (5.5)$$

$$= \left( \frac{-\Omega^2 + \epsilon}{b\Omega} \right)^2 + \frac{1}{b^2}. \quad (5.6)$$

Upon rearrangement, I am left with an equation quadratic in  $\Omega^2$  given by

$$\Omega^4 + \Omega^2(1 - b^2 - 2\epsilon) + \epsilon^2 = 0, \quad (5.7)$$

which can be solved for  $\Omega^2$  using the quadratic equation

$$\Omega^2(b) = \frac{1}{2} \left( b^2 + 2\epsilon - 1 \pm \sqrt{(b^2 + 2\epsilon - 1)^2 - 4\epsilon^2} \right). \quad (5.8)$$

Only one of these solutions (the one corresponding to the positive sign) actually solves the original stability equations, which can be determined numerically by substituting Eq. 5.8 back into Eqs. 5.3 and 5.4. Thus, we are left with two solutions

$$\Omega(b) = \pm \sqrt{\frac{1}{2} \left( b^2 + 2\epsilon - 1 \pm \sqrt{(b^2 + 2\epsilon - 1)^2 - 4\epsilon^2} \right)}, \quad (5.9)$$

which can be substituted into Eq. 5.4 to determine  $b$ . Since the equation is transcendental,  $b$  can take on multiple values for a given  $\tau$  and  $\epsilon$ . I am interested in the case when the fixed point loses stability as  $\gamma$  increases from zero; therefore, the only solutions of interest are the positive and negative solutions with the smallest magnitude,

$b_+$  and  $b_-$ . For the values of  $\epsilon$  and  $\tau$  that correspond to my experiment, I obtain  $b_+ \approx 1$  and  $b_- \approx -1$ .

Recall that  $b$  is the effective slope the nonlinearity, which depends on  $\gamma$  and  $m$ . One can thus determine the value of  $\gamma_H$  where the fixed point undergoes a Hopf bifurcation as a function of  $m$

$$\gamma_H = -\frac{b_{\pm}}{d \sin(2m)}. \quad (5.10)$$

Since  $\gamma$  is a nonnegative quantity in the experiment, I take  $b_+$  for  $m < 0$  and  $b_-$  for  $m > 0$ . Notice that, if  $b_+ \neq b_-$ , which can occur for various combinations of  $\tau$  and  $\epsilon$ , the Hopf curves are asymmetric about  $m = 0$ . For  $m = 0$ , which corresponds to the operating point at the top of the interference fringe,  $\gamma_H$  diverges and the fixed point of the noise-free model is linearly stable for all values of  $\gamma$ , as shown in Figs. 4.14 and 4.16.

## 5.2 Numerical Simulations

To numerically simulate solutions for my system, I use a multi-step predictor-corrector method known as the four-point Adams-Bashforth-Moulton (ABM) method, which is described in Ref. [37]. Consider the equation  $x' = f(x, t)$ , so that  $x(t) = s_n + \int_{t_n}^t f(x, t) dt$ . The ABM method involves three steps: prediction, evaluation, and correction. In the predictor step, a preliminary value for  $x_{n+1}$  is determined from

$$x_{n+1} = x_n + \frac{h}{24} (55x'_n - 59x'_{n-1} + 37x'_{n-2} - 9x'_{n-3}). \quad (5.11)$$

This value of  $x_{n+1}$  is then used to compute  $x'_{n+1}$  in the evaluation step. Finally,  $x_{n+1}$  is recomputed in the corrector step using the value of  $x'_{n+1}$  according to

$$x_{n+1} = x_n + \frac{h}{24} (9x'_n + 19x'_{n-1} - 5x'_{n-2} + x'_{n-3}). \quad (5.12)$$

In order to easily compare the numerical results with the experiment, I integrated the stochastic physical model. Solving Eq. 2.70 for  $V'/\Delta$  and defining

$$U = \Delta\epsilon \int_{t_0}^t V(l)dl - G \cos^2 m, \quad (5.13)$$

yields the system of equations used in the integration procedure

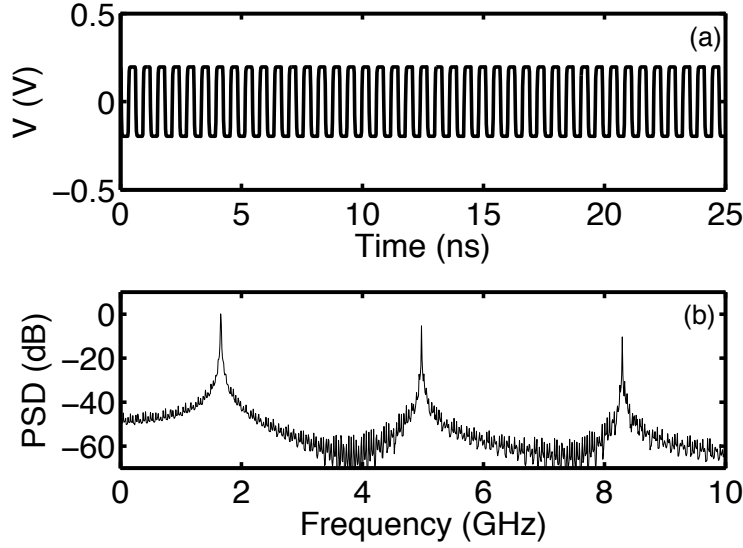
$$\frac{V'(t)}{\Delta} = -V(t) - U(t) + G \cos^2 \{m + d \tanh [gV(t - T)]\} - G \cos^2 m + N(t), \quad (5.14)$$

$$\frac{U'(t)}{\Delta} = \epsilon V(t). \quad (5.15)$$

The solutions obtained via numerical integration will depend on the parameters values as well as the initial conditions supplied. Since Eq. 5.14 depends on  $V(t - T)$ , one needs to define  $V$  from  $t = -T$  to  $t = 0$  and  $U$  at  $t = 0$  to start the integration process.

To check that the model and integration routine are correct, I determine the value of  $\gamma$  for which the steady-state loses stability at  $m = \pm\pi/4$ . I initialize the system in the steady-state ( $V = U = 0$ ) and use a uniformly distributed random variable centered around zero with a maximum amplitude of 5 mV for  $N(t)$ . For both values of  $m$ , I find that  $\gamma_{th,sim} \approx 0.49$ , which is in exact agreement with the theoretical value obtained from linear stability analysis ( $\gamma_H = 0.4878$ ) to the desired precision.

After ensuring the integration routine is functioning properly, I integrate the model for fixed  $\gamma$  in the traditional operating point regime. Using a sinusoidal wave with an

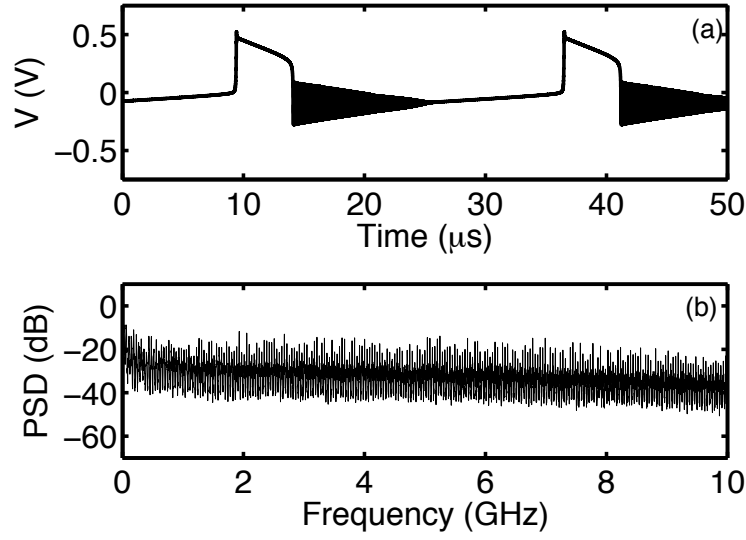


**Figure 5.1:** Periodic solution obtained from numerical integration with parameter values  $m = -\pi/4$  and  $\gamma = 0.94$ .

amplitude of 10 mV as an initial condition for the parameter values  $m = -\pi/4$  and  $\gamma = 0.94$ , I obtain the periodic solution shown in Fig. 5.1. In addition, integrating from the steady state with parameter values of  $m = -0.23$  and  $\gamma = 1.3$ , I find a breather solution shown in Fig. 5.2. These solutions should be compared to the experimental dynamics obtained at the same operating points shown in Figs. 4.8 and 4.9.

I am also able to reproduce the broadband chaotic behavior seen in my experiments for  $m = 0$  and  $\gamma = 4.36$ , using a Gaussian pulse with an amplitude of 65 mV and a FWHM of approximately 0.2 ns as the initial condition for  $V$ . As one can see from Fig. 5.3, the initial transient is breather-like before the chaotic behavior takes over at around 20  $\mu s$ . Like in the experimental case, the power spectrum is flat (contained within an 8 dB window with a standard deviation of 1.5 over the 0-8 GHz range), but without the 8 GHz roll-off due to the oscilloscope, as shown in Fig. 5.4.

Finally, for  $m = 0$ , I determine the critical input pulse amplitude  $V_{th,sim}$  for which the

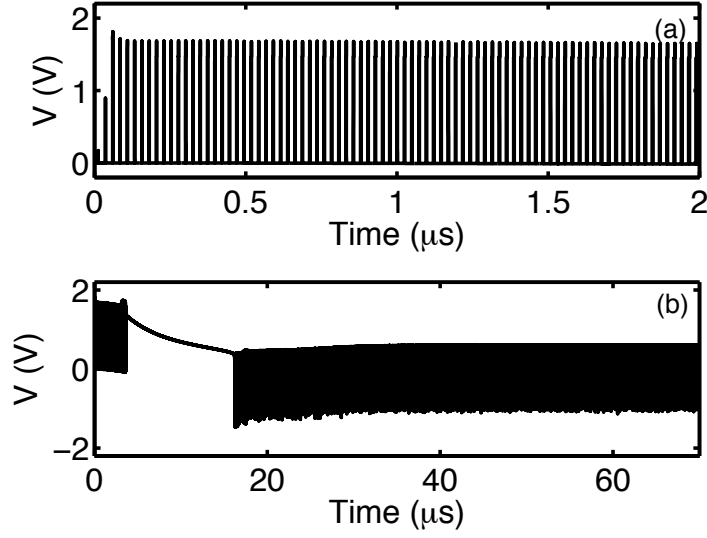


**Figure 5.2:** Breathers solution obtained from numerical integration with parameter values  $m = -0.23$  and  $\gamma = 1.3$ .

amplitude envelope of the subsequent pulse train will first start to grow as a function of  $\gamma$ . To do this, I again use a Gaussian pulse with a FWHM of approximately 0.2 ns as my initial condition. By varying the amplitude of the initial pulse and integrating for six or seven round trip times, I find the values of  $V_{th,sim}$  depicted in Fig. 5.5.

### 5.3 Phase Portrait Analysis

As previously reported in Chapter 4, I see experimentally that, for  $m = 0$ , the fixed point solution transitions into a broadband chaotic solution as  $\gamma$  increases. Furthermore, the dynamics during the transition away from the steady-state behavior are shown to be a sequence of narrow pulses with increasing amplitudes separated in time by  $T$ , as illustrated in Fig. 4.17 and 5.3. I also see this behavior numerically when I use a pulse with an amplitude greater than  $V_{th,sim}$  as the initial condition for  $V$ .



**Figure 5.3:** The transient behavior that occurs when the fixed point first loses stability for parameter values  $m = 0$  and  $\gamma = 4.36$  upon numerical integration of Eqs. 5.14 and 5.15. The initial pulse train highlighted in (a) leads into the breather-like behavior shown in (b).

One way to understand how this type of transition may arise is to consider the phase portrait for

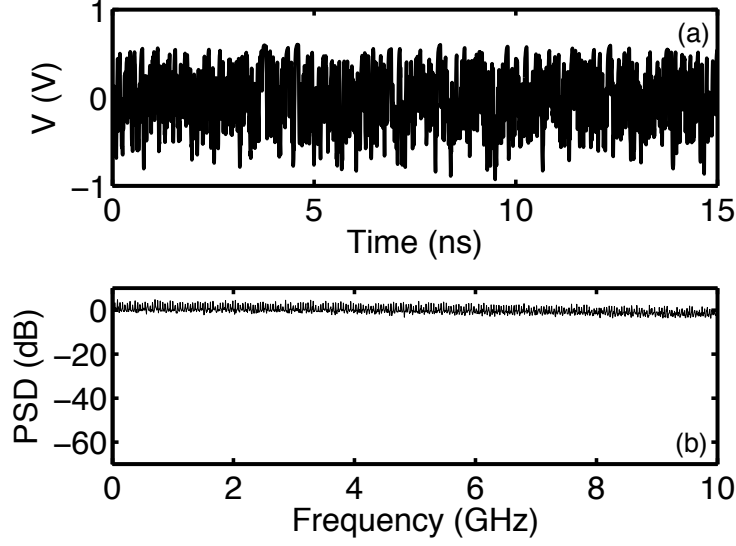
$$\dot{x} = -x(s) - y(s) + c[x(s - \tau)], \quad (5.16)$$

$$\dot{y} = \epsilon x(s), \quad (5.17)$$

where,  $c[x] = \gamma \cos^2(d \tanh x) - \gamma$  is the nonlinear delayed-feedback term from before with  $m = 0$ . When  $x(s - \tau)$  is zero,  $c[x(s - \tau)]$  vanishes and one can treat this as a two-dimensional system. Solving for the nullclines of Eqs. 5.16 and 5.17 gives

$$\dot{x} = 0 \implies y = -x, \quad (5.18)$$

$$\dot{y} = 0 \implies x = 0, \quad (5.19)$$



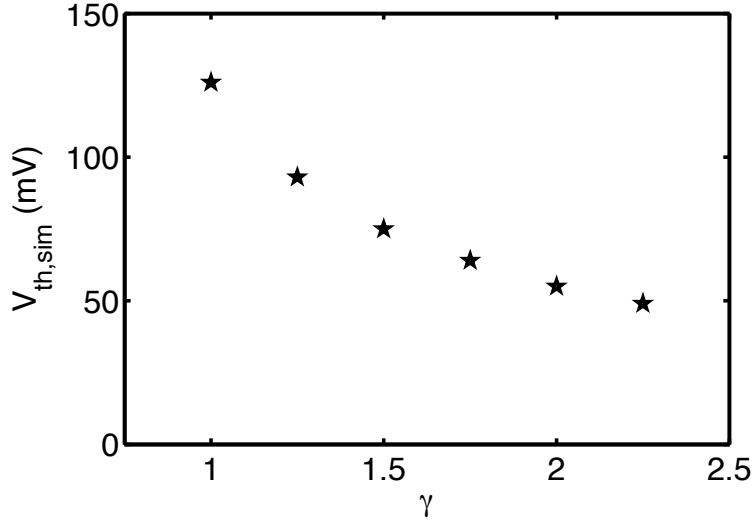
**Figure 5.4:** Time series (a) and power spectrum (b) of the broadband chaos obtained from a numerical simulation with  $m = 0$  and  $\gamma = 4.36$ .

Standard analysis techniques will show that the fixed point is a stable node, and that trajectories are drawn to the origin along the  $y = -x$  nullcline, as the small parameter  $\epsilon$  makes the motion parallel to the  $y$ -axis slow compared to motion parallel to the  $x$ -axis.

Now consider what happens when a short pulse with amplitude  $-x_0$  and centered at time  $s = 0$  occurs in the dynamic variable  $x$  (possibly due to a noise glitch) and is allowed to propagate through the feedback loop. Near time  $\tau$ , the feedback term begins to grow from zero, corresponding to the growth of the pulse a time  $\tau$  earlier. This will gradually shift the nullcline of Eq. 5.18 and, hence, the fixed point. When the feedback term reaches its maximum value at time  $s = \tau$ , the the nullcline is given by

$$y = -x + \gamma \cos^2(d \tanh x_0) - \gamma, \quad (5.20)$$

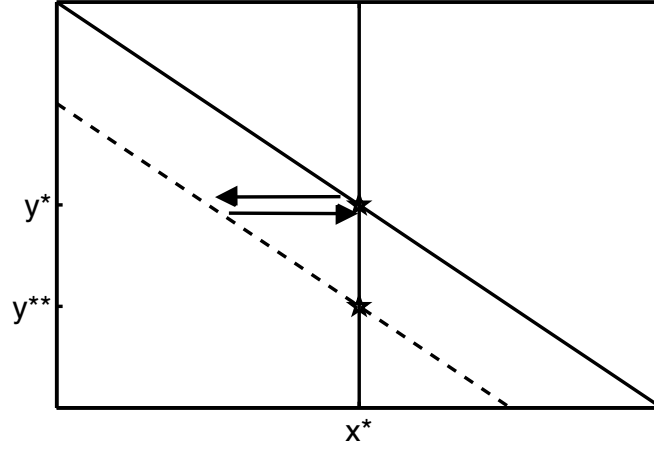
and the fixed point is now located at  $x^* = 0$ ,  $y^{**} = \gamma \cos^2(d \tanh x_0) - \gamma$ . As time



**Figure 5.5:**  $V_{th,sim}$  as a function of  $\gamma$ .

continues to increase, this nullcline gradually shifts back to its original location. The process is illustrated in Fig. 5.6.

Now consider what happens to trajectories that start near the stable node as the nullcline shifts over time. At times prior to  $\tau$ , these trajectories are drawn to the origin. For times near  $\tau$ , however, these trajectories are drawn towards the continually shifting nullcline, returning to the origin a short time later as the nullcline shifts back. Approximately, this produces another pulse in the dynamic variable  $x$ , again with a negative amplitude. Note that original pulse could have either a positive or negative amplitude for this to occur. Thus, this second pulse at time  $\tau$  will generate a third pulse at time  $2\tau$ , which will generate a fourth pulse at time  $3\tau$ , and so on, all with negative amplitudes. This is consistent with the experimental transient behavior for  $m = 0$  shown in Figs. 4.17 and 5.3, keeping in mind that  $V$  and  $x$  are inversely proportional to one another.



**Figure 5.6:** The nullclines with and without the presence of a pulse. The  $x = 0$  nullcline remains unchanged under the influence of a pulse, but the nullcline at  $y = -x$  (line) is shifted to  $y = -x - y^{**}$  (dashed) when the pulse reaches its maximum amplitude. Trajectories originating near  $(x^*, y^*)$  approximately follow the shifting nullcline, but are unable to reach  $(x^*, y^{**})$  since motion parallel to the  $y$ -axis is slow.

## 5.4 One-Dimensional Map

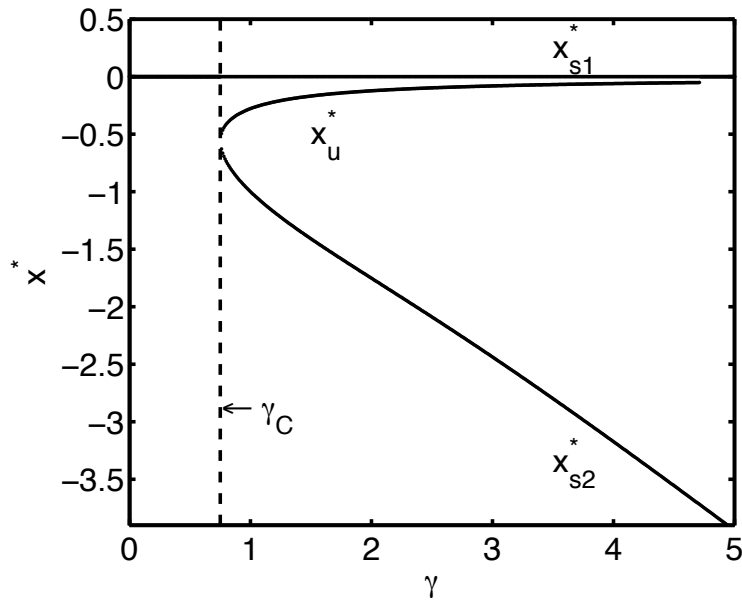
The phase-portrait analysis given above explains how the system can produce equally spaced pulses with negative amplitudes if first seeded with a pulse. This serves as motivation to investigate a one-dimensional map of the form

$$x_{n+1} = \gamma \cos^2[m + d \tanh(x_n)] - \gamma \cos^2 m, \quad (5.21)$$

$$\equiv f(x_n), \quad (5.22)$$

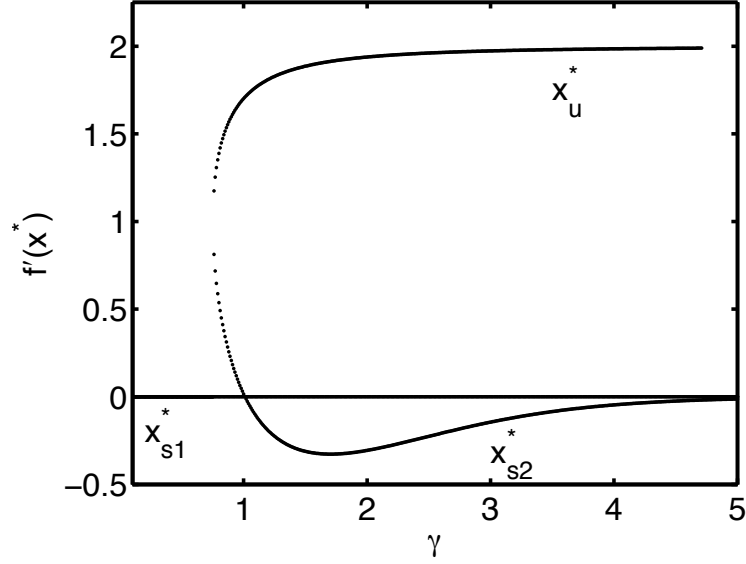
where  $x_n$  can be thought of as the amplitude of a pulse at time  $n\tau$  and the slowly changing variable  $y$  has been neglected. One should keep in mind, however, that the map given by Eq. 5.21 should only approximately predict the dynamics of the physical system, as reducing the DDE to a map erases all of the effects of the bandpass filter.

At  $m = 0$  there can be one or three fixed points, depending on the value of  $\gamma$ , as shown in Fig. 5.7. As the equation to determine the fixed points is transcendental, I calculate the fixed points numerically in Matlab. I compute the stability of each fixed point by evaluating the derivative of the map with respect to  $x_n$  at the corresponding fixed point (shown in Fig. 5.8) and find that the fixed point at the origin ( $x_{s1}^*$ ) is always stable. When the other two fixed points exist, they are both negative. However, the fixed point with the smaller magnitude ( $x_u^*$ ) is unstable, while the fixed point with the greater magnitude ( $x_{s2}^*$ ) is stable.



**Figure 5.7:** Fixed points of the one-dimensional map derived to approximate the system's pulsing behavior. For  $\gamma > \gamma_C$  there are three fixed points ( $x_{s1}^*$ ,  $x_u^*$ , and  $x_{s2}^*$ ).

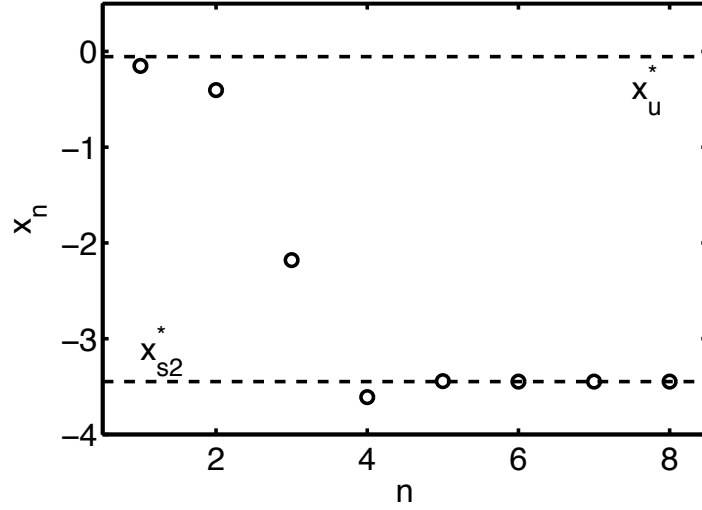
Setting  $\gamma = 4.36$ , which corresponds to the value of  $\gamma$  where the transient behavior is recorded experimentally, I iterate the map using the height of the first pulse seen experimentally as the initial condition. As one can see in Fig. 5.9, subsequent pulses grow in amplitude until they reach the fixed point at  $x_n = -3.45$ . This is analogous to the pulsing behavior that I observe experimentally, where a large enough input pulse



**Figure 5.8:** The first derivative of Eq. 5.21 evaluated at each fixed point as a function of  $\gamma$  with  $m = 0$ . A magnitude greater than one indicates that the fixed point is unstable.

(due to a noise glitch or an applied perturbation) creates a series of pulses separated in time by  $T$  with amplitudes that grow initially before saturating. I thus hypothesize that  $x_u^*$ , which is a function of  $\gamma$ , can be used to give an approximate criteria for the critical amplitude of a pulse needed to create a train of pulses with increasing amplitudes.

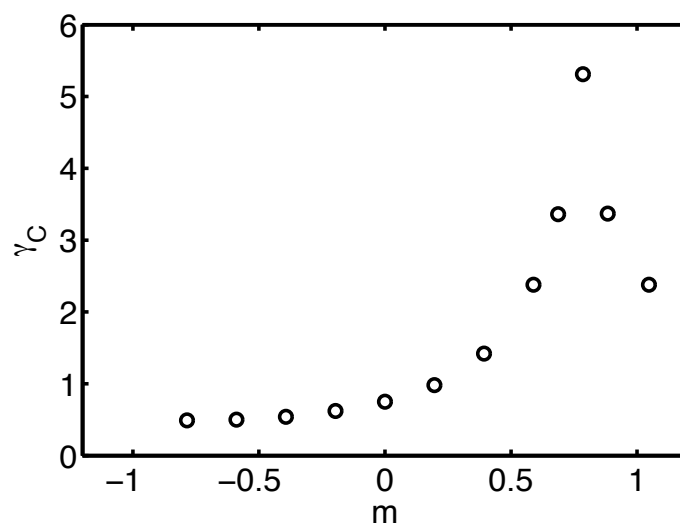
Another interesting feature of the map, as pointed out by Schoell, is that, for a given value of  $m$ , there exists a value of  $\gamma_C$  where the fixed points  $x_u^*$  and  $x_{s2}^*$  coalesce. One might hypothesize that this value would roughly correspond to the experimentally determined  $\gamma_{th-}$ , because, at this point, the origin is the only remaining fixed point. Using Mathematica, I determine  $\gamma_C$  for several values of  $m$ , as shown in Fig. 5.10.



**Figure 5.9:** Results from iterating map for  $\gamma = 4.36$  with  $x_1 = 10.152$ , which corresponds to the amplitude of the first pulse recorded in the experimental transient for the same  $\gamma$ . The fixed points  $x_u^*$  and  $x_{s2}^*$  are indicated with horizontal dashed lines.

## 5.5 Summary

In this chapter, I show that the steady-state at  $m = 0$  is linearly stable for all values of  $\gamma$ . Just as in my experiments, however, I find via numerical integration that the steady-state behavior at  $m = 0$  can transition to non-steady-state behavior with the input of a narrow pulse. This eventually leads to broadband chaotic dynamics with a flat power spectrum. I also determine, as a function of  $\gamma$ , the amplitude of an input pulse  $V_{th,sim}$  necessary to generate a train of pulses with increasing amplitude for  $m = 0$ . The idea that an input pulse can create a train of pulses is explained by analyzing the phase space of the dimensionless model. Finally, I explore the features of a map derived by neglecting the effects of the bandpass filter on the system, in hopes of gaining insight as to how the coexisting attractor for  $m = 0$  can be reached.



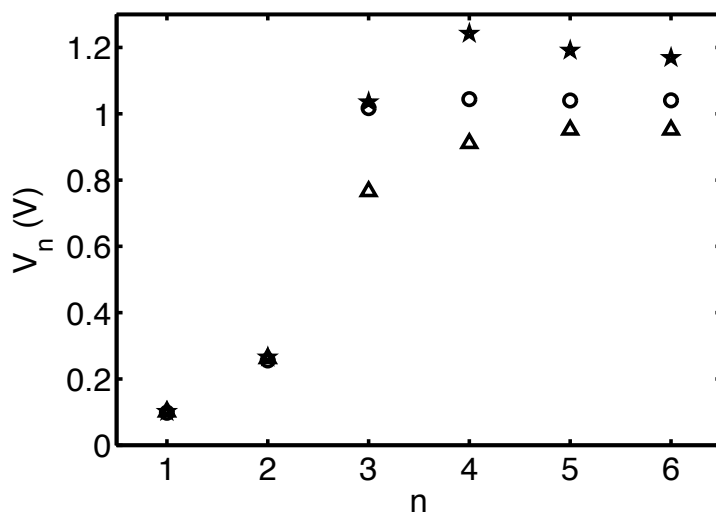
**Figure 5.10:** Numerical results for  $\gamma_C$ , the value of  $\gamma$  where  $x_u^*$  and  $x_{s2}^*$  collide, as a function of  $m$ .

# Chapter 6

## Conclusion

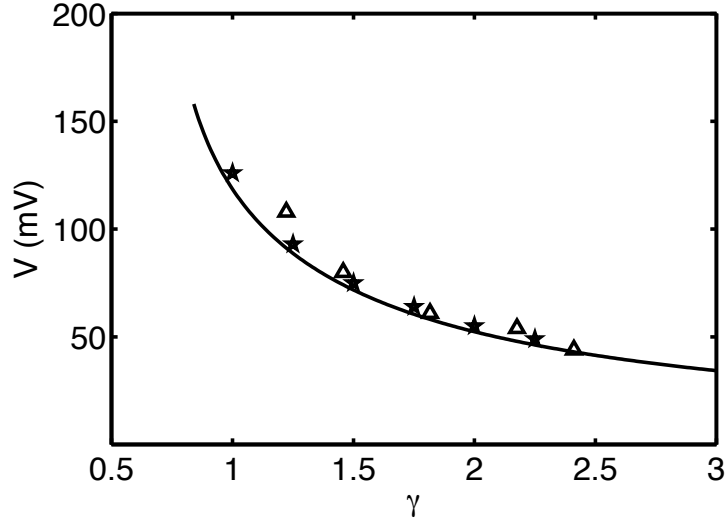
### 6.1 Comparison of Results

In the previous chapters, I investigate the behavior of a time-delayed opto-electronic oscillator via four avenues: experimentation with the physical system, linear stability analysis of a noise-free model, numerical simulation of a stochastic model, and analysis of a map derived to approximate certain features of the physical system and full model.



**Figure 6.1:** Comparison of pulse amplitudes obtained for  $m = 0$  and  $\gamma = 4.26$  in the experiment (triangles), simulation (stars), and map (circles).

The long-term dynamics for fixed  $m$  and  $\gamma$  achieved in the experiment and simulation are found to be similar, both at the traditional and new operating points. Most

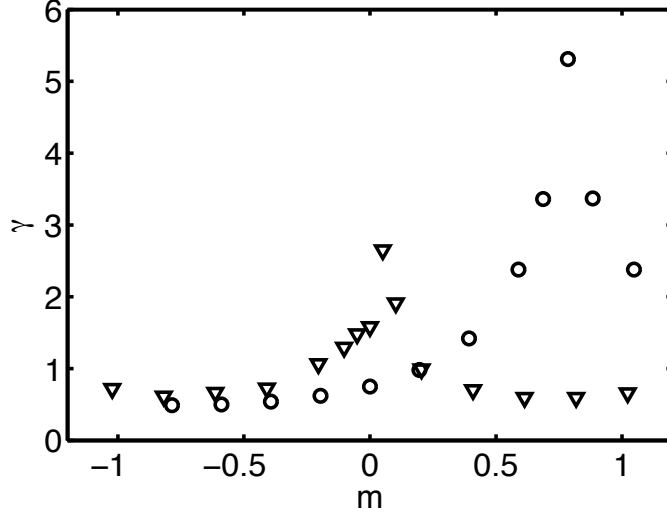


**Figure 6.2:** Comparison of  $V^*$  (line),  $V_{th,sim}$  (stars), and  $V_{th,exp}$  (triangles).

notably, the broadband chaos recorded experimentally for  $m = 0$  was reproduced numerically using an short pulse as the initial condition for  $V$ . This exciting result has yet to be reported in the literature and could imply the existence of a type of attractor with a dense set of unstable periodic orbits, each with a similar stability.

The amplitude growth of the transient pulse train for  $m = 0$  and  $\gamma = 4.36$  recorded experimentally is also qualitatively reproduced with both the simulation and the map, as shown in Fig. 6.1. Furthermore, I find excellent agreement between  $V_{th,exp}$ ,  $V_{th,sim}$ , and  $V^*$ , as shown in Fig. 6.2, where  $V^* = g * x_u^*$ . This indicates that the condition  $V = V^*$  is a valid criteria for when a perturbation in the form of a short pulse will grant access to the broadband chaotic attractor.

As shown in Fig. 4.14, my experimental determination of  $\gamma_{th+}$  and  $\gamma_{th-}$  is asymmetric about  $m = 0$ . In an effort to explain the asymmetry in  $\gamma_{th-}$ , I calculate  $\gamma_C$ , the value of  $\gamma$  where the two nonzero fixed points of the map coalesce. This quantity is also asymmetric about  $m = 0$ , however, the quantitative agreement between  $\gamma_{th-}$  and



**Figure 6.3:** Comparison of  $\gamma_{th-}$  (upside-down triangles) and  $\gamma_C$  (circles).

$\gamma_C$  is poor, as can be seen in Fig. 6.3.

## 6.2 Future Directions

There are several interesting questions and ideas that can be pursued with this system. First, an analysis of the stochastic model (Eq. 2.70) could provide additional insight about the conditions under which the coexisting chaotic attractor can be accessed. Dramatically altering the bandpass characteristics with the addition of a narrow filter into the feedback loop, as Larger *et al.* did in [32], is another area where an analysis of a stochastic model could provide additional insight about the stability of an RF oscillation.

In addition to studying its stochastic nature, one could also investigate the effect of the time delay further. For example, some interesting dynamics have come about by networking oscillators with a non-negligible time-delay in the coupling. In particular,

Fischer *et al.* found that by coupling three oscillators in a line, with the middle device bidirectionally coupled to both outer devices, the outer two oscillators could synchronize with no time lag, while the middle oscillator was also synchronized but with a time lag equal to the time delay induced by the coupling [38]. Interestingly enough, there exists an operating regime where the middle oscillator leads the outer two and a regime where the middle oscillator lags the other two. An experiment like this one would be easy to reproduce with the opto-electronic device described here by coupling three systems electronically or optically, which could open the way for discovering new types of network dynamics.

# Appendix A

## Mathematical Definitions

In this appendix I present some mathematical definitions useful for describing noise characteristics. The discussion is similar to that given by Boyd in [16].

Assume a noise signal  $\delta X(t)$  is only nonzero for times in the range  $-T/2$  and  $T/2$ . Its Fourier transform is given by

$$\delta \tilde{X}(\omega) = \int_{-T/2}^{T/2} \delta X(t) e^{-i\omega t} dt \quad (\text{A.1})$$

The spectral density is then defined to be

$$S(\omega) = \lim_{T \rightarrow \infty} \left\langle \frac{1}{T} |\delta \tilde{X}(\omega)|^2 \right\rangle, \quad (\text{A.2})$$

where the angled brackets denote an ensemble average. Since ensemble averages are difficult to measure, a more practical definition of the spectral density is given by

$$S(\omega) = \lim_{T \rightarrow \infty} \frac{1}{T} |\delta \tilde{X}(\omega)|^2, \quad (\text{A.3})$$

which usually converges to the real value for physical systems. Note that if  $\delta X^2$  is proportional to power (which is typically the case), then the spectral density is proportional to the power per unit frequency interval.

Using Parseval's theorem

$$\int_{-\infty}^{\infty} |\delta X(t)|^2 dt = \int_{-\infty}^{\infty} |\delta \tilde{X}(\omega)|^2 d\omega, \quad (\text{A.4})$$

one can show that

$$\overline{\delta X^2} = \int_{-\infty}^{\infty} S(\omega) d\omega, \quad (\text{A.5})$$

$$= 2 \int_0^{\infty} S(\omega) d\omega. \quad (\text{A.6})$$

In the last step I used the fact that since  $\delta X(t)$  is real,  $S(\omega)$  must be even. It is common to define the one-sided spectral density to be  $W(\omega) = 2S(\omega)$ , so that

$$\overline{\delta X^2} = \int_0^{\infty} W(\omega) d\omega. \quad (\text{A.7})$$

The autocorrelation of  $\delta X(t)$  is defined to be

$$C(\tau) = \overline{\delta X(t) \delta X(t + \tau)}, \quad (\text{A.8})$$

and one can show that this function has its maximum value at  $\tau = 0$  and is equal to

$$C(0) = \overline{\delta X^2(t)}. \quad (\text{A.9})$$

Finally, the Wiener-Khintchine theorem states that the spectral density and autocorrelation function form a Fourier transform pair. The Wiener-Khintchine relations are

$$W(\omega) = 4 \int_0^{\infty} C(\tau) \cos \omega \tau d\tau, \quad (\text{A.10})$$

$$C(\tau) = \frac{1}{2\pi} \int_0^{\infty} W(\omega) \cos(\omega \tau) d\omega. \quad (\text{A.11})$$

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